Linear Regression: Loss, Least Squares, and Projection

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia, Prof. Krishna Prasad Miyapuram Source: `Lectures/L04a_student_loss_risk_erm.pdf` (33 slides + 6 board pages = 39 pages) and `Lectures/L04b_student_linear_regression_four_routes.pdf` (24 slides, solved classroom copy) · Previous: Estimating Parameters

How to use this file. Part A builds the *new* maths Lecture 4 needs, on top of Lectures 1–3 (likelihood, logs, MLE, derivatives, argmin). Most of it is linear algebra: vectors, matrices, column space, rank, gradients of matrix expressions. Part B goes through every page of 4A (including the handwritten board pages) and every slide of 4B. The "work it out" slides in 4A are left blank in the PDF, so they are **fully solved** here, with numbers on the running dataset. Part C is practice.

What this lecture is really about: Lecture 3's recipe, applied to a prediction that *depends on an input*:

$$\underbrace{Y_i \mid x_i \sim \mathcal{N}(\theta_0 + \theta_1 x_i,\ \sigma^2)}_{\text{model}} \xrightarrow{-\log \prod} \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{least squares}} \xrightarrow{\nabla = 0} \underbrace{X^\top X \hat{\theta} = X^\top y}_{\text{normal equations}} \longrightarrow \underbrace{\hat{y} = P_X\, y}_{\text{projection}}$$

Change the noise model and the loss changes (Laplace → absolute error). Change nothing and look at it geometrically: least squares is **the nearest point in the column space**, and the residual is **perpendicular** to it.

Step-up path

You need. Dot products, matrix shapes, and the MLE recipe.

1. See residuals as errors, and squared loss as the area of squares.
2. Learn where the loss comes from: Gaussian noise gives squares, and Laplace noise gives absolute values.
3. Stack everything into X and y, and write the loss as a single norm.
4. Differentiate to get the normal equations.
5. Read those equations as geometry: the residual is perpendicular to every column.

You can stop when you can derive $X^\top X\theta = X^\top y$ and draw what it means.

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Part A — Maths from the ground up

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1. Projection matrices

Part B — Slides · 4A (39 pages) · 4B (24 slides)

Part C — Practice · Jump

Part A Maths From the Ground Up

A1 Expectation and conditional expectation

Expectation = probability-weighted average (L1 §8):

$$\mathbb{E}[X] = \sum_x x\,p(x) \quad \text{(discrete)}, \qquad \mathbb{E}[X] = \int x\,p(x)\,dx \quad \text{(continuous)}.$$

Board analogy (civil engineering). Loads F_1, F_2, F_3 sit on a beam at distances x_1, x_2, x_3 from end A. The moment about A is $\sum_k F_k x_k$. Replace the loads by probability masses $p(x_k)$ and the moment becomes $\sum x\,p(x) = \mathbb{E}[X]$: the **mean is the balance point** (centre of mass) of the distribution. Higher **moments** follow the same pattern: $\mathbb{E}[X^2] = \sum x^2 p(x)$, ..., $\mathbb{E}[X^k]$.

Rules you need:

Rule	Meaning
$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b$	constants pull out (linearity)
$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$	always, even if dependent
$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$	second moment minus squared mean

Conditional expectation $\mathbb{E}[Y \mid X = x]$: the average of Y *among cases where the input equals* x. It's a **function of** x, and it's the thing regression tries to estimate.

Board derivation. Suppose $Y = wX + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ independent of X. Then

$$\mathbb{E}[Y \mid X] = \mathbb{E}[wX \mid X] + \mathbb{E}[\varepsilon] = wX + 0 = wX.$$

Knowing X makes wX a constant; the noise averages to zero. So **the line is the conditional mean**, and the data scatter around it with spread σ.

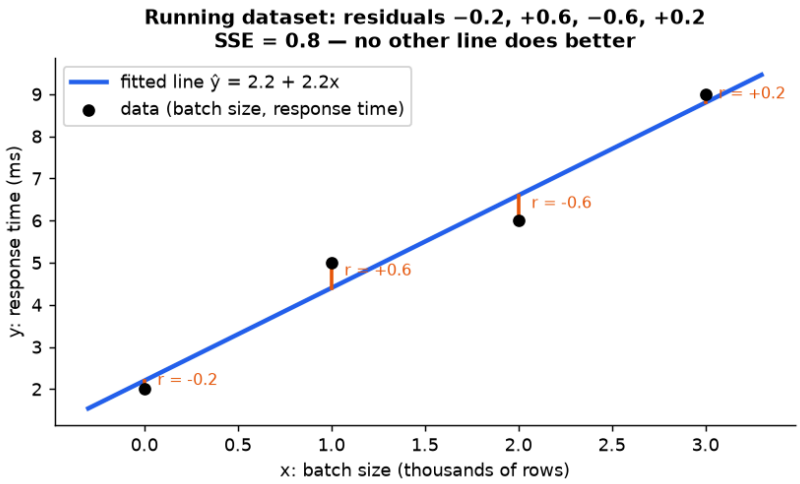
A2 Five objects: parameter → prediction

L3 §A1 separated data from parameters. Lecture 4 splits the parameter side further:

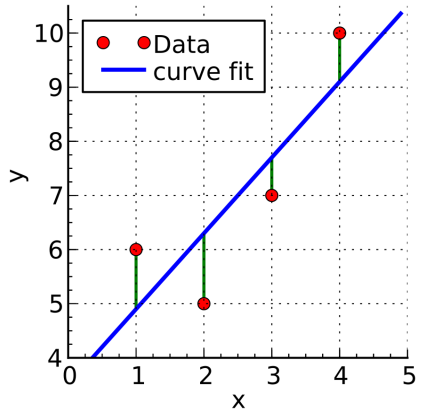
Object	Notation	What it is	Running example
parameter	θ	unknown true value inside the model	true intercept and slope
estimator	$\hat{\theta}(\mathcal{D})$	a *rule* that maps any dataset to a value	$(X^\top X)^{-1}X^\top y$
estimate	$\hat{\theta}$	the number the rule gives on *this* data	$(2.2,\ 2.2)$
fitted model	$f_{\hat{\theta}}$	the function with the estimate plugged in	$f(x) = 2.2 + 2.2x$
prediction	$\hat{y}_\star = f_{\hat{\theta}}(x_\star)$	output for one chosen input	$x_\star = 4 \Rightarrow 11.0$

An optimizer doesn't "learn a dataset". It **applies an estimator** to the dataset and returns an estimate.

A3 Residuals, losses and risk



The 4A running dataset with its least-squares line 2.2 + 2.2x and every residual labelled.



Residuals are the vertical gaps (green) between data points and the fitted line. (Wikimedia Commons: Linear_least_squares_example2.svg, CC BY-SA 3.0)

For a candidate θ and row i:

$$\hat{y}_i = f_\theta(x_i), \qquad r_i(\theta) = y_i - \hat{y}_i \quad \text{(residual: keeps its sign)}.$$

A **loss** $\ell(y, \hat{y})$ turns one miss into a non-negative cost:

Loss	$\ell(r)$	Character
squared	r^2	big errors cost a lot more (doubling the error ×4 the cost)
absolute	$\lvert r \rvert$	every unit of error costs the same
Huber ($\delta > 0$)	$\frac{1}{2}r^2$ if $\lvert r \rvert \le \delta$; $\delta(\lvert r \rvert - \frac{1}{2}\delta)$ otherwise	quadratic near 0, linear in the tails
asymmetric	e.g. $a\,r$ if $r > 0$, $b\lvert r \rvert$ if $r < 0$	under- and over-prediction cost differently

(Check Huber is continuous at $\lvert r \rvert = \delta$: $\frac{1}{2}\delta^2$ vs $\delta(\delta - \frac{1}{2}\delta) = \frac{1}{2}\delta^2$ ✓. The slopes also match: δ on both sides, so it's smooth.)

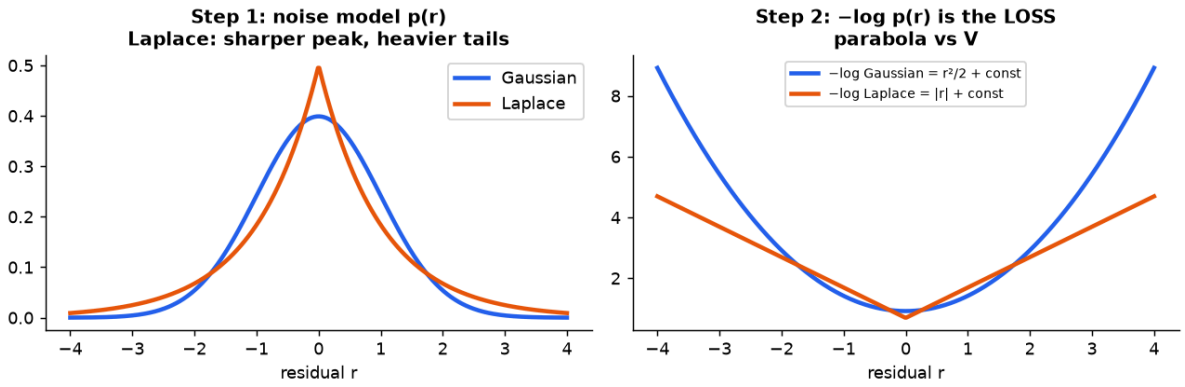
From one loss to an objective:

Name	Formula	Can we compute it?
empirical risk	$\hat{R}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, f_\theta(x_i))$	yes: training data
population risk	$R(\theta) = \mathbb{E}_{(X,Y)\sim P}\big[\ell(Y, f_\theta(X))\big]$	no: needs the unknown P

Empirical risk minimisation (ERM): $\hat{\theta} \in \arg\min_\theta \hat{R}_n(\theta)$. We minimise what we can compute, hoping it tracks what we care about (L7–L8 is about when it doesn't).

($\in$ rather than $=$ because the argmin can be a *set*; see the flat valley, A10.)

A4 Negative log-likelihood is a loss generator



Pick a noise curve, take –log: the Gaussian gives a parabola (squared loss), the Laplace a V (absolute loss).

L3's MLE, with an input:

$$\hat{\theta}_{\text{MLE}} = \arg\max_\theta \prod_{i=1}^{n} p(y_i \mid x_i, \theta) = \arg\min_\theta \sum_{i=1}^{n} \underbrace{-\log p(y_i \mid x_i, \theta)}_{\text{loss of row } i}.$$

(Allowed because $-\log$ is strictly *decreasing*: it flips max into min without moving the location.)

The translation trick (4B slide 5). Write $m_i = f_\theta(x_i)$ and $y_i = m_i + \varepsilon_i$. Sliding the axis by the fixed number m_i doesn't change heights, so

$$p_{Y|X}(y_i \mid x_i, \theta) = p_\varepsilon(y_i - m_i) = p_\varepsilon(r_i(\theta)).$$

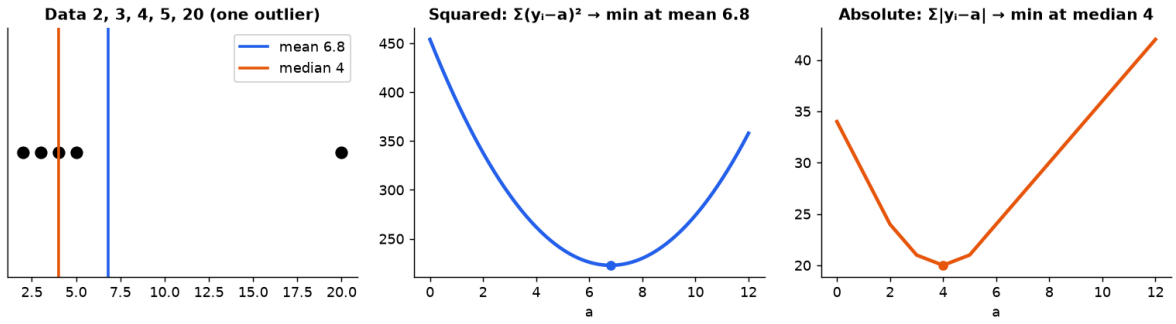
The noise density, evaluated at the residual, is the likelihood. So $\ell(r) = -\log p_\varepsilon(r)$:

Noise ε	Density	$-\log p_\varepsilon(r)$	Loss
Gaussian $\mathcal{N}(0, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-r^2/2\sigma^2}$	$\frac{1}{2}\log(2\pi\sigma^2) + \frac{r^2}{2\sigma^2}$	**squared**
Laplace$(0, b)$	$\frac{1}{2b} e^{-\lvert r\rvert/b}$	$\log(2b) + \frac{\lvert r\rvert}{b}$	**absolute**

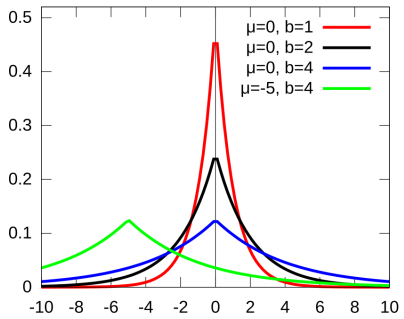
Constants and positive scale factors don't move the argmin, so they are dropped.

Laplace has a **sharper peak and heavier tails** than the Gaussian: big errors are less surprising, so they are penalised linearly instead of quadratically.

A5 Squared loss → mean, absolute loss → median



Data 2, 3, 4, 5, 20: squared loss bottoms out at the mean 6.8, absolute loss at the median 4.



Laplace PDF: sharp peak and heavier tails than the normal; its NLL is the absolute loss. (Wikimedia Commons: Laplace_pdf_mod.svg, CC BY-SA 4.0)

Fit one constant a to data $y_1..y_n$.

Squared: $J_2(a) = \sum (y_i - a)^2$. $J_2'(a) = -2\sum (y_i - a) = 0 \Rightarrow a = \bar{y}$ **(mean)**.

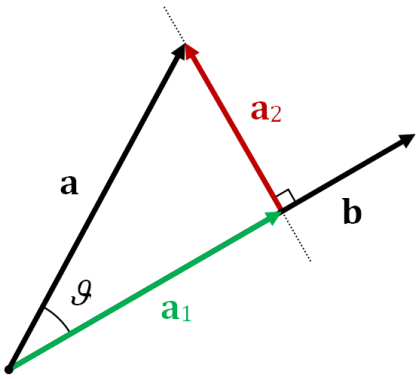
Absolute: $J_1(a) = \sum |y_i - a|$. Away from data points, $\frac{d}{da}|y_i - a| = -1$ if $y_i > a$ and $+1$ if $y_i < a$, so

$$J_1'(a) = \#\{y_i < a\} - \#\{y_i > a\}.$$

It's zero when as many points are below a as above: $a =$ **median**.

The same holds in the population, conditionally on x: the minimiser of $\mathbb{E}[(Y-a)^2 \mid x]$ is $\mathbb{E}[Y \mid x]$; of $\mathbb{E}[|Y-a| \mid x]$ it's the conditional median. **The loss decides which summary of $Y \mid x$ you are estimating.**

A6 Vectors: dot product, norm, orthogonality



Projection a_1 of a onto b, and the perpendicular remainder a_2 (the 'residual'). (Wikimedia Commons: Projection_and_rejection.png, CC BY-SA 3.0)

A vector $v \in \mathbb{R}^n$ is a column of n numbers. Stack the n responses into $y \in \mathbb{R}^n$ and the n residuals into $r \in \mathbb{R}^n$.

Operation	Formula	Note
dot product	$u^\top v = \sum_i u_i v_i$	a scalar; $u^\top v = v^\top u$
squared length	$\|v\|_2^2 = v^\top v = \sum_i v_i^2$	lower 2 = Euclidean; upper 2 = squared
orthogonal	$u \perp v \iff u^\top v = 0$	right angle
expand a square	$\|u - v\|^2 = \|u\|^2 - 2u^\top v + \|v\|^2$	the cross term vanishes if $u \perp v$ (Pythagoras)

So **SSE is a squared length**: $\sum_i r_i^2 = \|r\|_2^2 = \|y - X\theta\|_2^2$.

A7 Matrices: shapes, transpose rules, design matrix

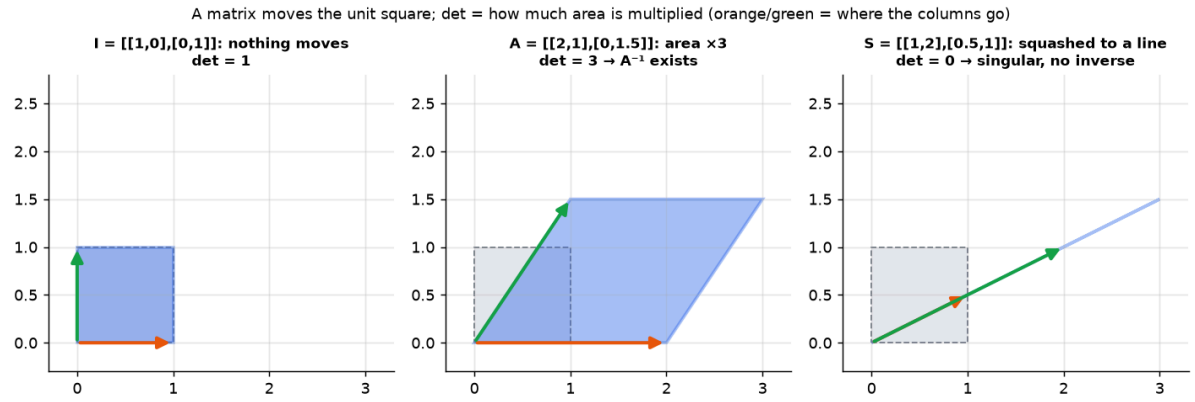
$A \in \mathbb{R}^{m\times k}$ times $B \in \mathbb{R}^{k\times p}$ gives $\mathbb{R}^{m\times p}$. **Inner sizes must match.** Always check shapes (L2's "dimensions as a correctness check").

Transpose rules: $(A^\top)^\top = A$, $(A+B)^\top = A^\top + B^\top$, $(AB)^\top = B^\top A^\top$ (order flips), $(A^{-1})^\top = (A^\top)^{-1}$.

A scalar equals its own transpose. If $y^\top X\theta$ is 1×1, then $y^\top X\theta = (y^\top X\theta)^\top = \theta^\top X^\top y$. This merges the two middle terms when you expand $\|y - X\theta\|^2$.

Symmetric: $A^\top = A$. $X^\top X$ is always symmetric: $(X^\top X)^\top = X^\top X$.

Matrix toolkit: identity, inverse, determinant, singular



Identity leaves the square alone; det 3 triples its area; det 0 flattens it, and a flattened square can't be un-flattened.

These four words appear in every formula from here on (normal equations, projection, eigenvalues), so build them one step at a time. Picture a 2×2 matrix as a machine that **moves every point of the plane**. Its first column says where the arrow $(1,0)$ lands; its second column says where $(0,1)$ lands.

Step 1 — Identity I: the "do nothing" matrix. $I = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$. Check: $I\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}x\\y\end{bmatrix}$. It plays the role of the number 1: $AI = IA = A$. (I_d or I_n just says its size.)

Step 2 — Inverse A^{-1}: the "undo" matrix. The number 5 has the inverse $\frac{1}{5}$ because $5\times\frac{1}{5} = 1$. Same idea: $A^{-1}A = AA^{-1} = I$. If A moves a point, A^{-1} moves it back. That's how we "divide" by a matrix: from $A\theta = c$, multiply both sides on the left by A^{-1} to get $\theta = A^{-1}c$.

Step 3 — Determinant $\det A$: the area multiplier. Send the unit square (area 1) through A. It becomes a parallelogram with area $|\det A|$. For 2×2, $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.

Matrix	Where the square goes	det
$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	stays put	1
$\begin{bmatrix} 2 & 1 \\ 0 & 1.5 \end{bmatrix}$	parallelogram, 3× the area	$2(1.5) - 1(0) = 3$
$\begin{bmatrix} 1 & 2 \\ 0.5 & 1 \end{bmatrix}$	flattened onto a line	$1(1) - 2(0.5) = 0$

Step 4 — Singular means det = 0, and then no inverse exists. If the square is squashed onto a line, many different points land on the same spot. You can't tell which one you started from, so there's nothing to "undo". Notice the singular example's second column $(2, 1)$ is just $2\times$ the first $(1, 0.5)$: **dependent columns ⇔ det = 0 ⇔ no inverse**. That is exactly the rank problem in A8 and slide 31.

2×2 inverse formula: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (swap the diagonal, negate the off-diagonal, divide by det). It exists iff $ad - bc \neq 0$, and the division by det shows why.

> **Check:** $A = \begin{bmatrix} 2 & 1 \\ 0 & 1.5 \end{bmatrix}$. $A^{-1} = \frac{1}{3} \begin{bmatrix} 1.5 & -1 \\ 0 & 2 \end{bmatrix}$. Multiply: $AA^{-1} = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = I$ ✓.

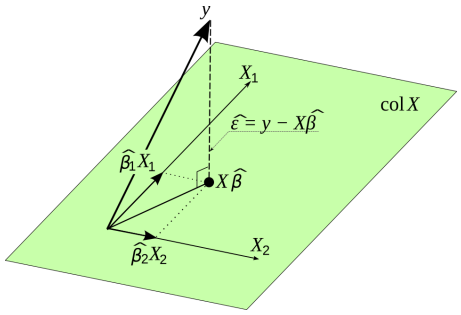
Design matrix. Choose features $\phi(x) = [\phi_0(x), \ldots, \phi_{d-1}(x)]^\top$ (e.g. $[1, x]$ or $[1, x, x^2]$). Then $\hat{y}_i = \phi(x_i)^\top \theta$, and stacking the rows $\phi(x_i)^\top$ gives

$$X = \begin{bmatrix} \phi(x_1)^\top \\ \vdots \\ \phi(x_n)^\top \end{bmatrix} \in \mathbb{R}^{n \times d}, \qquad \hat{y} = X\theta \in \mathbb{R}^n, \qquad r = y - X\theta.$$

Row i = one observation. **Column** j = one feature measured across all observations. The column of 1s is what carries the intercept.

"Linear" regression means linear in θ, not in x. $\theta_0 + \theta_1 x + \theta_2 x^2$ is a curve in x but still linear regression: each θ_j appears to the first power, multiplying a fixed function. θ_1^2, $\theta_1\theta_2$, e^{θ_1} would not be.

A8 Span, column space and rank



OLS geometry: ŷ = Xβ̂ is the foot of the perpendicular from y onto the column space of X. (Wikimedia Commons: OLS_geometric_interpretation.svg, Public domain)

Write $X = [c_0\ c_1\ \cdots]$ by columns. Then

$$X\theta = \theta_0 c_0 + \theta_1 c_1 + \cdots$$

is a **linear combination** of the columns. The columns are fixed; the coefficients vary.

- **Span** $\{c_0, c_1\}$ = every vector you can make as $a\,c_0 + b\,c_1$.
- **Column space** $\mathrm{Col}(X) = \{X\theta : \theta \in \mathbb{R}^d\} \subseteq \mathbb{R}^n$ = every prediction vector the model can produce. The observed y usually isn't in it (otherwise the fit would be perfect).
- **Rank** $= \dim \mathrm{Col}(X)$ = number of genuinely independent column directions.
- **Full column rank** ($\mathrm{rank}(X) = d$): no column is a combination of the others. Equivalently, $Xv = 0 \Rightarrow v = 0$, i.e. the **null space** $\mathrm{Null}(X) = \{0\}$.
- **Rank deficient** ($\mathrm{rank}(X) < d$): some $v \neq 0$ has $Xv = 0$. Then $X(\theta + tv) = X\theta$ for every t: different parameters, same predictions.

Example: $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ has two equal columns, so rank 1, and $v = (1, -1)$ is in the null space.

A9 Gradients of matrix expressions

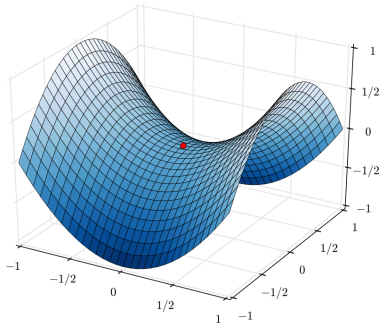
For a scalar function $J(\theta)$ of a vector $\theta \in \mathbb{R}^d$, the **gradient** $\nabla_\theta J \in \mathbb{R}^d$ stacks the partial derivatives (L2 §A6). It has the **same shape as** θ.

$J(\theta)$	$\nabla_\theta J$	Scalar analogue
$b^\top\theta$ (or $\theta^\top b$)	b	$\frac{d}{d\theta} b\theta = b$
$\theta^\top A\theta$	$(A + A^\top)\theta$	$\frac{d}{d\theta} a\theta^2 = 2a\theta$
$\theta^\top A\theta$, A symmetric	$2A\theta$	
$\frac{1}{2}\|y - X\theta\|^2$	$X^\top(X\theta - y) = -X^\top r$	$\frac{d}{d\theta}\frac{1}{2}(y - x\theta)^2 = -x(y - x\theta)$

Why $(A + A^\top)\theta$ (4B slide 23): $\theta^\top A\theta = \sum_j \sum_k \theta_j A_{jk} \theta_k$. θ_m appears as the first factor ($j = m$) and as the second ($k = m$), so $\partial/\partial\theta_m = \sum_k A_{mk}\theta_k + \sum_j \theta_j A_{jm} = [(A + A^\top)\theta]_m$.

Differentials check (4B slide 24): $r = y - X\theta$, $dr = -X\, d\theta$, $dJ = r^\top dr = -r^\top X\, d\theta = (-X^\top r)^\top d\theta$. Since $dJ = (\nabla J)^\top d\theta$, $\nabla J = -X^\top r$ ✓.

A10 Hessian, positive (semi)definite, convexity



Not convex: a saddle curves up one way and down the other (Hessian has a negative eigenvalue). (Wikimedia Commons: Saddle_point.svg, CC BY 3.0)

The **Hessian** $\nabla^2 J$ is the $d \times d$ matrix of second derivatives. For least squares,

$$\nabla^2 J = \nabla_\theta\left[X^\top X\theta - X^\top y\right] = X^\top X.$$

A symmetric matrix H is

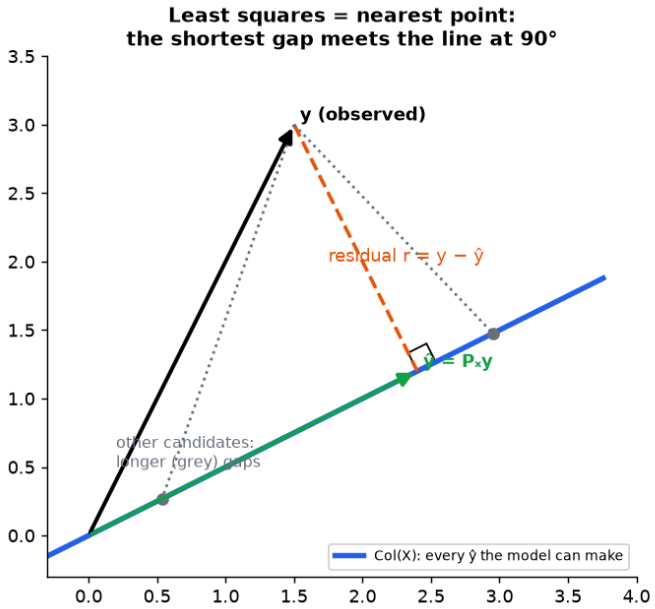
	Condition	Picture
positive semidefinite (PSD), $H \succeq 0$	$v^\top H v \geq 0$ for all v	bowl, possibly with a flat valley
positive definite (PD), $H \succ 0$	$v^\top H v > 0$ for all $v \neq 0$	strict bowl, single bottom point

For a twice-differentiable J: Hessian PSD everywhere ⇒ **convex**; PD everywhere ⇒ **strictly convex** ⇒ at most one minimiser. Convex ⇒ any point with $\nabla J = 0$ is a **global** minimum.

The key identity: $v^\top X^\top X v = (Xv)^\top (Xv) = \|Xv\|^2 \geq 0$. So least squares is **always convex in** θ. It's strictly convex exactly when $Xv \neq 0$ for every $v \neq 0$, i.e. **full column rank** (A8).

Careful. Always name the variable. J is convex **in** θ with X, y fixed. "The problem is convex" alone says nothing.

A11 Projection matrices



The nearest point on the line to y is where the gap is perpendicular; every other candidate is farther.

A square matrix P is an **orthogonal projection** if

1. **idempotent**: $P^2 = P$ (projecting twice changes nothing);
2. **symmetric**: $P^\top = P$ (makes the leftover perpendicular).

Then for any y: Py is the "shadow" of y on the subspace, and $(I - P)y$ is the perpendicular leftover. $(I - P)$ is also a projection, onto the orthogonal complement, and $P(I - P) = P - P^2 = 0$.

For least squares (full column rank): $P_X = X(X^\top X)^{-1}X^\top$ (an $n \times n$ matrix). Useful fact: $\mathrm{trace}(P_X) = \mathrm{rank}(X) = d$.

Part B1 Page by Page (all 39 pages)

The PDF footers count 33 slides; six handwritten board pages (pp. 7–12) were inserted after slide 6. Headings below give the **slide number from the footer**, with board pages labelled separately.

Running dataset for 4A (batch size x_i in thousands of rows, response time y_i in ms):

i	1	2	3	4
x_i	0	1	2	3
y_i	2	5	6	9

Slide 1 Title

Lecture 4A: Linear Regression I — From likelihood to least squares, projection, and uniqueness. Prof. Udit Bhatia, IIT Gandhinagar, Semester I 2026.

Slide 2 Course Roadmap: You Are Here

Still in **L3–L6 estimation and linear models**. Today's chain: conditional model → fitting objective → parameter estimate → geometric projection.

Slide 3 A Question From The Previous Class

Q: Why can a modern language model write complex software yet fail on a simple-looking number sequence?

- "Complex" isn't one mathematical quantity. Familiar code has strong patterns in the training data.
- A short sequence can support many **incompatible rules** (1, 2, 4, … could continue 8 or 7), and the prompt may not say which one generated it.

So a model still needs a **criterion**: which family of explanations is allowed, what counts as a costly error, which parameter value is preferred.

> **Insight:** today makes those three choices explicit: **model family, loss, estimator**.

Slide 4 Vocabulary We Will Reuse Today

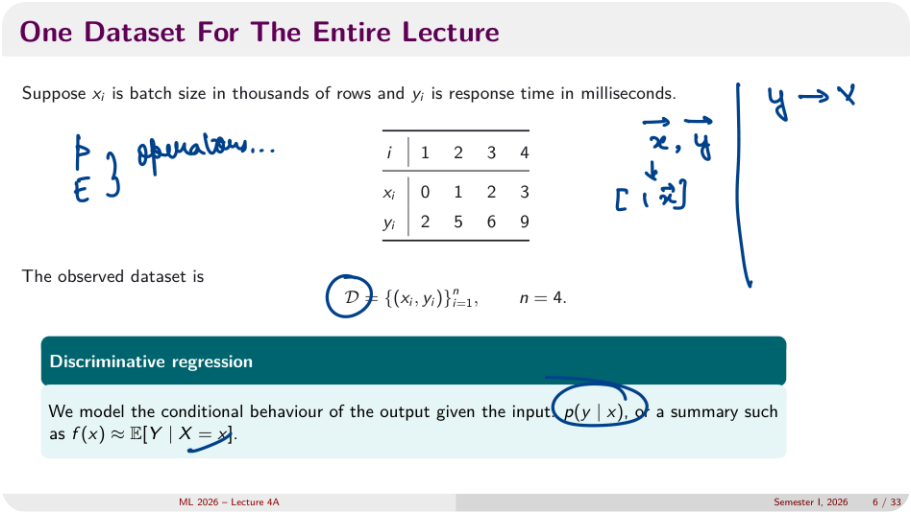
Idea	Meaning here
conditional probability	a distribution for the output once the input is known: $p(y \mid x)$
non-stationarity	the data-generating mechanism may change with time or context
weights / parameters	unknown numbers θ that pick one member of a model family
optimization	a rule for choosing θ, usually by minimising an objective
convexity	a property of the objective **as a function of a named variable**

Careful. Never say "the problem is convex" without naming the variable: convex in x, in θ, or in something else? (A10)

Slide 5 Learning Objectives

1. Distinguish parameter, estimator, estimate, fitted model, prediction (A2).
2. Derive squared-error fitting from a Gaussian conditional model (slides 14–16).
3. Explain why another noise model gives another loss (slide 21).
4. Derive the normal equations and the projection matrix (slides 25–26).
5. Prove the residual is orthogonal to the design space (slide 28).
6. Test whether least squares is convex in θ, and state when its minimiser is unique (slides 29–32).

Slide 6 One Dataset For The Entire Lecture



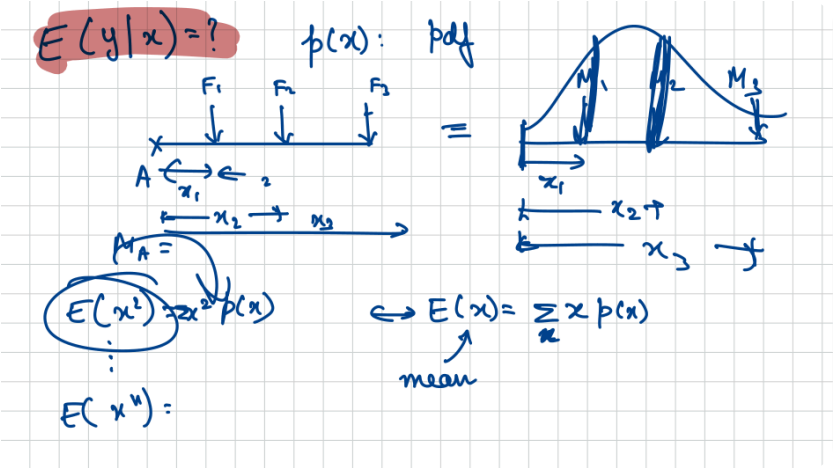
4A slide 6 — The four-point dataset used all lecture; teacher's ink marks y vs x.

The table above: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$, $n = 4$.

Discriminative regression: we model the output *given* the input, $p(y \mid x)$, or a summary of it such as $f(x) \approx \mathbb{E}[Y \mid X = x]$. We don't model how the x's themselves are distributed.

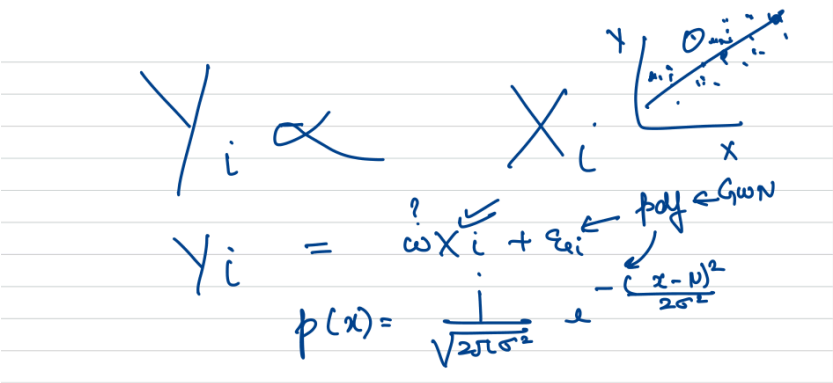
Board pp. 7–12 — What was written in class

p. 7 — What is $\mathbb{E}[y \mid x]$? Beam with loads F_1, F_2, F_3 at distances x_1, x_2, x_3 from A; moment $M_A = \sum F_k x_k$. Same picture with a PDF: the masses are $p(x)$, so $\mathbb{E}[X] = \sum_x x\, p(x)$ is the **mean** (first moment); $\mathbb{E}[X^2] = \sum x^2 p(x)$, …, $\mathbb{E}[X^n]$ are higher moments. (A1)



Board p. 7 — Beam with loads: E[X] is a 'moment', each value weighted by its probability.

p. 8 — The model. Scatter of Y vs X roughly along a line, so $Y_i \propto X_i$. Write

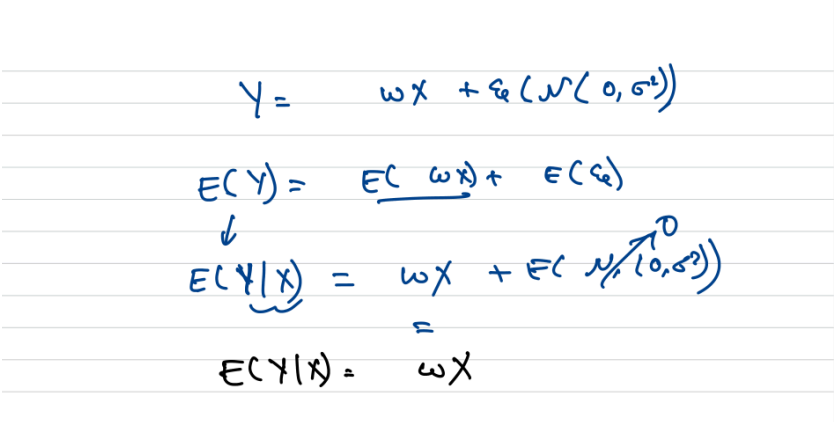


Board p. 8 — Points roughly on a line: $Y_i = wX_i + \varepsilon_i$ with Gaussian noise.

$$Y_i = wX_i + \varepsilon_i,$$

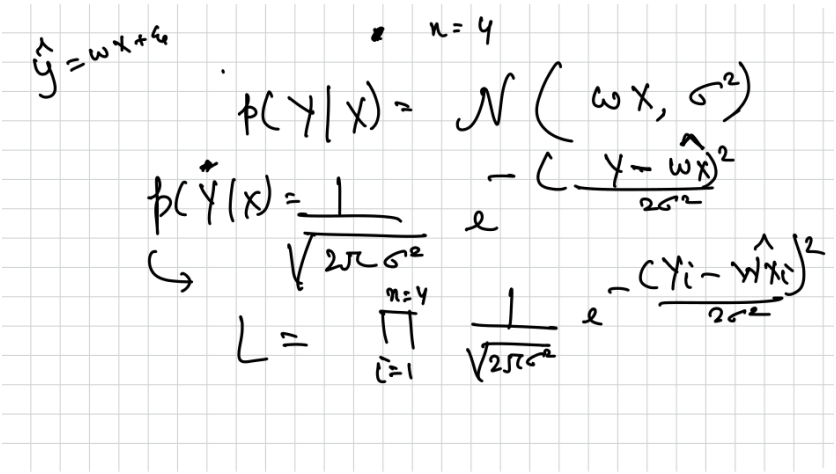
with ε_i having a PDF: Gaussian white noise (**GWN**), $p(\varepsilon) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(\varepsilon-\mu)^2/2\sigma^2}$ with $\mu = 0$. X_i is known; w is the unknown.

p. 9 — Conditional mean. $Y = wX + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. Taking expectations given X: $\mathbb{E}[Y \mid X] = wX + \mathbb{E}[\varepsilon] = wX + 0 = wX$. The line is the conditional mean.



Board p. 9 — E[Y | X] = wX because the noise averages to 0.

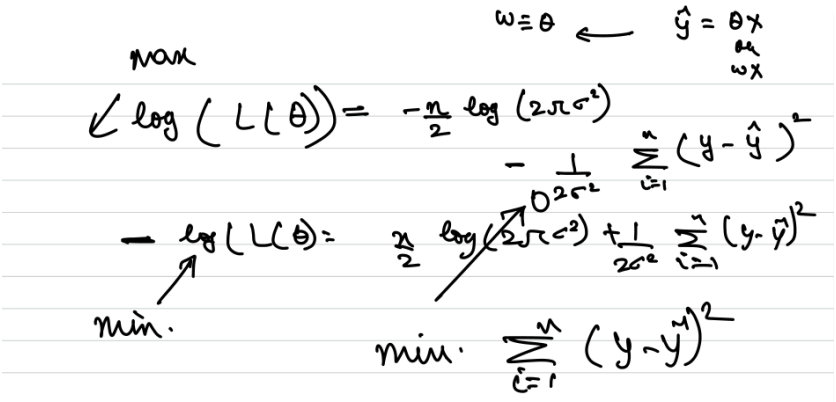
p. 10 — Likelihood for $n = 4$. $p(Y \mid X) = \mathcal{N}(wX, \sigma^2)$, so



Board p. 10 — Likelihood for n = 4 as a product of four bells.

$$p(y_i \mid x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y_i - wx_i)^2/2\sigma^2}, \qquad L = \prod_{i=1}^{4} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y_i - wx_i)^2/2\sigma^2}.$$

p. 11 — Max likelihood = min squared error. With $w \equiv \theta$, $\hat{y} = \theta x$:

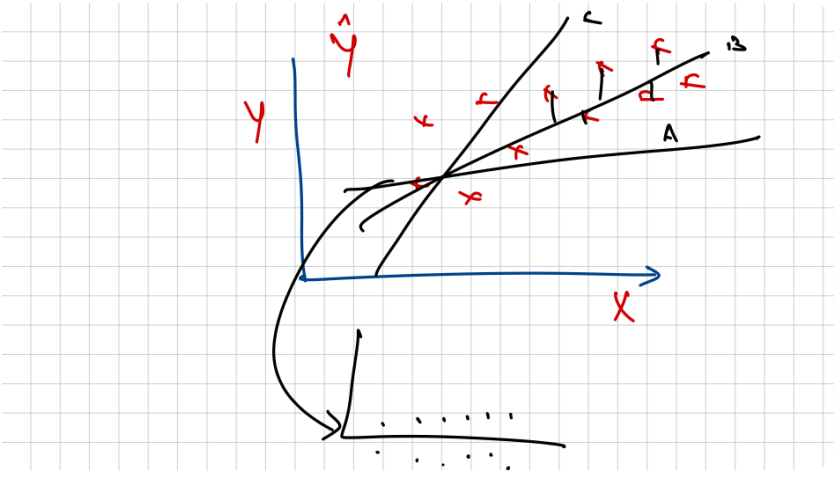


Board p. 11 — Max log L ⇔ min Σ(y − ŷ)².

$$\log L(\theta) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

Maximise $\log L$ ⇔ minimise $-\log L = \frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum(y_i - \hat{y}_i)^2$. The first term has no θ (its derivative is 0), so: $\min_\theta \sum_i (y_i - \hat{y}_i)^2$.

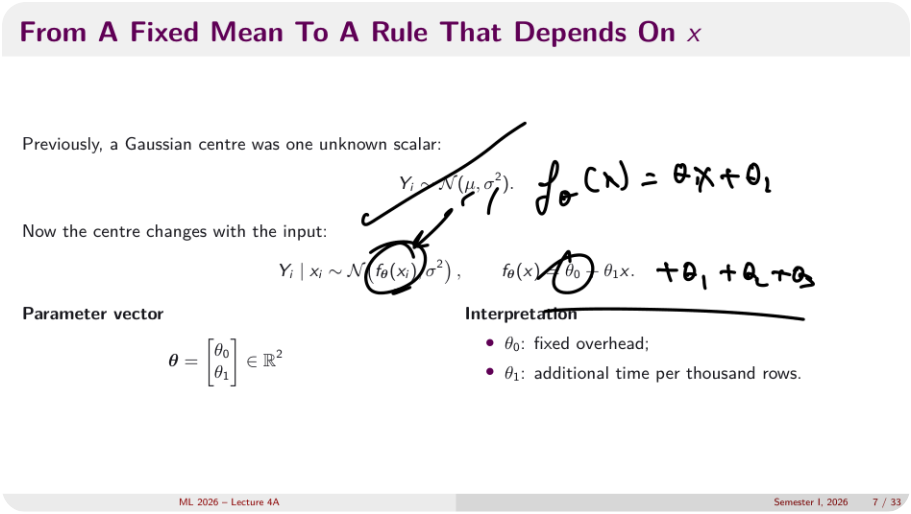
p. 12 — Which line? Three candidate lines A, B, C through one point of the data cloud. For each, the **residuals** r are the vertical gaps from data points to the line. Plotting residuals against x for a good line should show a structureless band around 0 (this is how you check the noise assumption, slide 18).



Board p. 12 — Three candidate lines A, B, C; red ticks are residuals.

Slide 7

From A Fixed Mean To A Rule That Depends On x



4A slide 7 — The centre now moves with x: μ becomes $\theta_0 + \theta_1 x$.

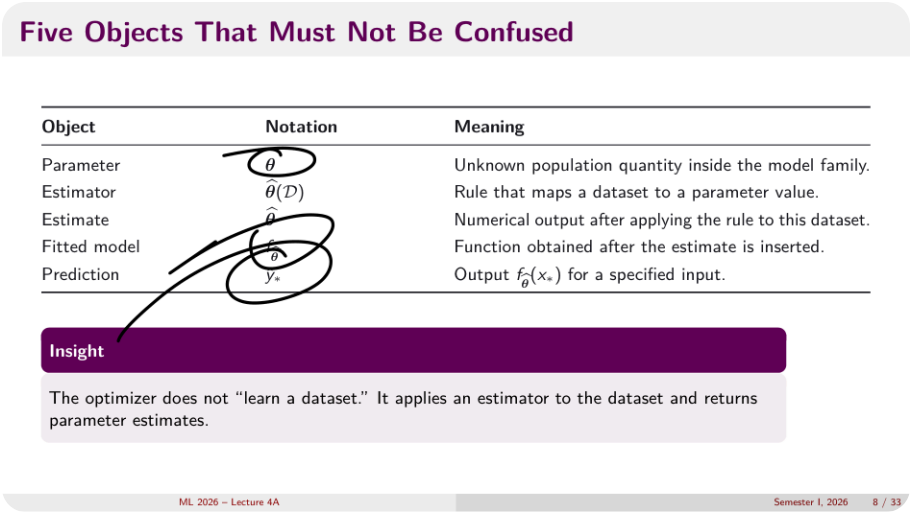
L3: one unknown centre, $Y_i \sim \mathcal{N}(\mu, \sigma^2)$. Now the centre moves with the input:

$$Y_i \mid x_i \sim \mathcal{N}\big(f_\theta(x_i),\, \sigma^2\big), \qquad f_\theta(x) = \theta_0 + \theta_1 x, \qquad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix} \in \mathbb{R}^2.$$

- θ_0: fixed overhead (response time for an empty batch);
- θ_1: extra time per thousand rows.

Insight: L3's μ has become a *function* $f_\theta(x)$. Set $\theta_1 = 0$ and you're back to L3 ($\hat{\theta}_0 = \bar{y}$).

Slide 8 Five Objects That Must Not Be Confused



Five Objects That Must Not Be Confused

Object	Notation	Meaning
Parameter	θ	Unknown population quantity inside the model family.
Estimator	$\widehat{\theta}(\mathcal{D})$	Rule that maps a dataset to a parameter value.
Estimate	$\widehat{\theta}$	Numerical output after applying the rule to this dataset.
Fitted model	$f_{\widehat{\theta}}$	Function obtained after the estimate is inserted.
Prediction	$\widehat{y}_*$	Output $f_{\widehat{\theta}}(x_*)$ for a specified input.

Insight

The optimizer does not "learn a dataset." It applies an estimator to the dataset and returns parameter estimates.

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4A slide 8 — Parameter, estimator, estimate, fitted model, prediction.

See A2. Filled in on our data (using the answer from slide 25):

Object	Value
parameter θ	unknown
estimator	$\hat{\theta}(\mathcal{D}) = (X^\top X)^{-1} X^\top y$
estimate	$\hat{\theta} = (2.2,\ 2.2)$
fitted model	$f_{\hat{\theta}}(x) = 2.2 + 2.2x$
prediction at $x_\star = 4$	$\hat{y}_\star = 11.0$ ms

Insight: the optimiser applies an estimator to the dataset and returns parameter estimates.

Slide 9 A Residual Measures One Miss

Candidate line $f_\theta(x) = 1 + 1.5x$:

x_i	0	1	2	3
y_i	2	5	6	9
$\hat{y}_i$	1	2.5	4	5.5
$r_i = y_i - \hat{y}_i$	1	2.5	2	3.5

Residuals keep their sign (here all positive: the line is too low everywhere). A **loss** turns a residual into a non-negative cost. For this line SSE $= 1 + 6.25 + 4 + 12.25 = 23.5$. (The best line, slide 25, gets SSE $= 0.8$.)

Slide 10 The Data Do Not Choose The Loss

Loss	Formula	Emphasises
squared	r^2	large errors get rapidly increasing cost
absolute	$\lvert r \rvert$	each extra unit costs the same
Huber	quadratic then linear	smooth near 0, less sensitive in the tails
asymmetric	different cost by sign	under- and over-prediction needn't be equal

Q: is the loss a mathematical convenience, a probabilistic assumption, or a description of real decision cost? **It can be any of the three.** A professional analysis says which one it is using. (E.g. under-predicting flood height costs lives; over-predicting costs money, so asymmetric.)

Slide 11 From One Loss To An Objective

$$\hat{R}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, f_\theta(x_i)) \quad \text{(available)}, \qquad R(\theta) = \mathbb{E}_{(X,Y)\sim P}\big[\ell(Y, f_\theta(X))\big] \quad \text{(desired)}.$$

The first is computed from observed data; the second is performance on the generating population. (A3)

Slide 12 Parameter Fitting Is An Optimization Rule

ERM: $\hat{\theta} \in \arg\min_{\theta\in\Theta} \hat{R}_n(\theta)$.

model family f_θ → loss $\ell(y, \hat{y})$ → objective $\hat{R}_n(\theta)$ → estimate $\hat{\theta}$.

Careful. Model, objective, optimiser and evaluation metric are **different objects**. They may be chosen together, but they're not interchangeable. (E.g. fit by squared error, report RMSE, optimise by normal equations or gradient descent.)

Slide 13 Maximum Likelihood Is Also An Optimization Rule

With conditionally independent observations,

$$p(y \mid x, \theta) = \prod_{i=1}^{n} p(y_i \mid x_i, \theta), \qquad \hat{\theta}_{\text{MLE}} \in \arg\max_\theta p(y \mid x, \theta) = \arg\min_\theta \big[-\log p(y \mid x, \theta) \big],$$

because $-\log$ is strictly decreasing.

Insight: negative log-likelihood is a **loss-generating mechanism**. The assumed distribution decides the loss's shape. (A4)

Slide 14 One Gaussian Observation Contains A Square

$$p(y_i \mid x_i, \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \theta_0 - \theta_1 x_i)^2}{2\sigma^2} \right].$$

The residual $r_i(\theta) = y_i - \theta_0 - \theta_1 x_i$ sits **squared in the exponent**. Logs will bring it down.

Slide 15 From A Product To A Sum

$$\log L(\theta) = \log \prod_i p_i(\theta) = \sum_i \log p_i(\theta) = \sum_{i=1}^{n} \left[-\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{r_i(\theta)^2}{2\sigma^2} \right] = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n} r$$

Identical to L3 slide 16 with μ replaced by $f_\theta(x_i)$.

Slide 16 The Gaussian MLE Becomes Least Squares

Flip to minimisation: $\arg\max \log L = \arg\min(-\log L)$. With σ^2 fixed, the terms without θ form a constant C:

$$-\log L(\theta) = C + \frac{1}{2\sigma^2}\sum_{i=1}^{n}[y_i - f_\theta(x_i)]^2.$$

Drop C and the positive factor $\frac{1}{2\sigma^2}$:

$$\boxed{\hat{\theta}_{\text{MLE}} \in \arg\min_\theta \sum_{i=1}^{n}[y_i - f_\theta(x_i)]^2 = \arg\min_\theta \text{SSE}(\theta).}$$

Note σ^2 doesn't affect $\hat{\theta}$ at all (as in L3, where $\hat{\mu}$ didn't depend on σ).

Slide 17 SSE, MSE, And RMSE Fit The Same Parameters

$$\text{SSE} = \sum r_i^2, \qquad \text{MSE} = \tfrac{1}{n}\text{SSE}, \qquad \text{RMSE} = \sqrt{\text{MSE}}.$$

Multiplying by $1/n > 0$ and taking $\sqrt{\cdot}$ are strictly increasing, so all three have the **same argmin**.

Careful. The fitted parameters agree, but the reported numbers don't. RMSE is in the **units of** y (ms); MSE is in ms².

On our data at $\hat{\theta}$: SSE $= 0.8$, MSE $= 0.2$ ms², RMSE $= 0.447$ ms.

Slide 18 Noise And Residual Are Not The Same Object

$$Y_i = f_\theta(x_i) + \varepsilon_i.$$

Object	Notation	Meaning
noise / error	ε_i	random quantity in the data-generating model; never observed
residual	$r_i = y_i - f_{\hat{\theta}}(x_i)$	a number computed after fitting; depends on data and estimate

The distributional assumption is made about ε_i, not about residuals.

> **Insight:** residual plots are **evidence** for judging whether the assumed noise model is reasonable (board p. 12).

Slide 19 The Loss Can Change The Fitted Answer (solved)

Fit one constant a to $2, 3, 4, 5, 20$.

- **Squared:** $J_2'(a) = -2\sum(y_i - a) = 0 \Rightarrow a = \bar{y} = 34/5 = \mathbf{6.8}$.
- **Absolute:** minimiser is the median (A5) $= \mathbf{4}$. Check: at $a = 4$, two points below, two above, so $J_1' = 0$. $J_1(4) = 2 + 1 + 0 + 1 + 16 = 20$; $J_1(6.8) = 4.8 + 3.8 + 2.8 + 1.8 + 13.2 = 26.4$.

The **squared** loss is pulled much farther by the outlier 20: without it, the mean would be 3.5; with it, 6.8. The median only moves from 3.5 to 4.

Why: in the squared loss the outlier's pull is proportional to its residual ($2r$); in absolute loss every point pulls with strength 1 regardless of distance.

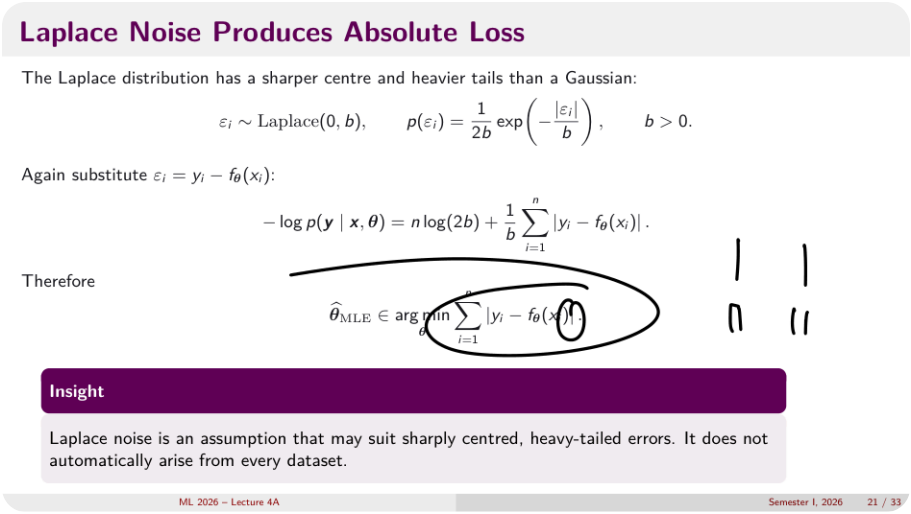
Slide 20 Gaussian Noise Produces Squared Loss

$\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. Substitute $\varepsilon_i = y_i - f_\theta(x_i)$, take $-\log$, sum:

$$-\log p(y \mid x, \theta) = \frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_i [y_i - f_\theta(x_i)]^2 \;\Rightarrow\; \hat{\theta}_{\text{MLE}} \in \arg\min_\theta \sum_i [y_i - f_\theta(x_i)]^2.$$

Slide 21

Laplace Noise Produces Absolute Loss



4A slide 21 — Laplace noise: –log turns |r| into the loss.

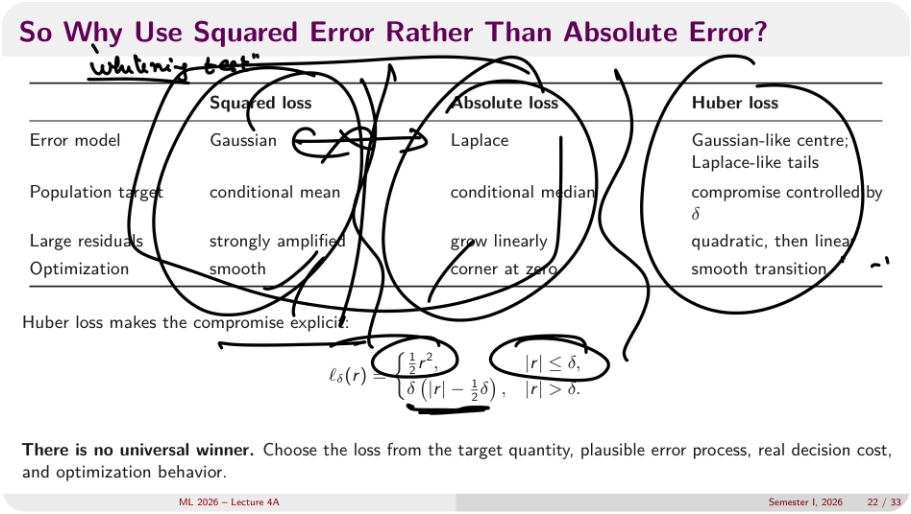
$\varepsilon_i \sim \text{Laplace}(0, b)$, $p(\varepsilon) = \frac{1}{2b}e^{-|\varepsilon|/b}$, $b > 0$. Sharper centre and heavier tails than a Gaussian.

$$-\log p(y \mid x, \theta) = n\log(2b) + \frac{1}{b}\sum_i |y_i - f_\theta(x_i)| \;\Rightarrow\; \hat{\theta}_{\text{MLE}} \in \arg\min_\theta \sum_i |y_i - f_\theta(x_i)|.$$

Insight: Laplace noise is an **assumption** that may suit sharply centred, heavy-tailed errors. It doesn't automatically hold for every dataset.

Slide 22

So Why Use Squared Error Rather Than Absolute Error?



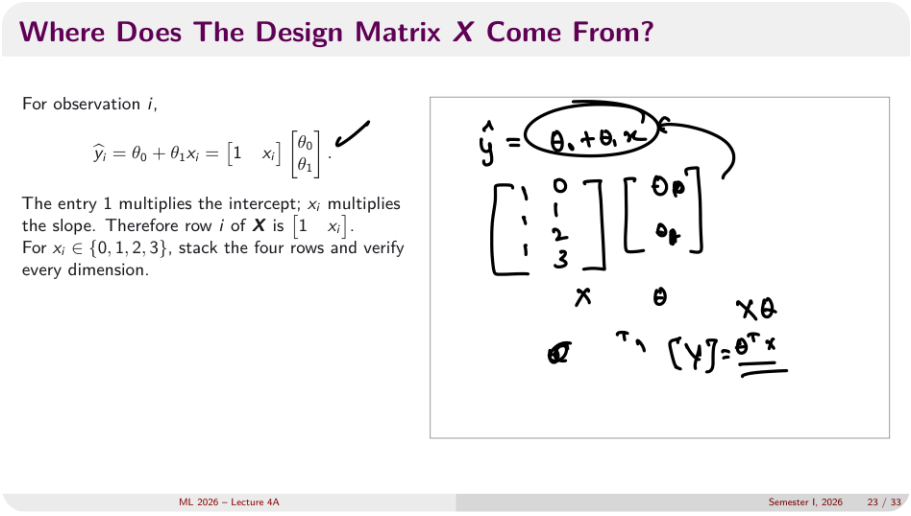
4A slide 22 — Ink sketches: parabola (squared), V (absolute), Huber (parabola then straight).

	Squared	Absolute	Huber
error model	Gaussian	Laplace	Gaussian-like centre, Laplace-like tails
population target	conditional **mean**	conditional **median**	compromise set by δ
large residuals	strongly amplified	grow linearly	quadratic, then linear
optimisation	smooth; closed form	corner at 0; no closed form	smooth transition

$$\ell_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta \\ \delta\left(|r| - \frac{1}{2}\delta\right), & |r| > \delta \end{cases}$$

No universal winner. Choose from: the target quantity, the plausible error process, the real decision cost, and optimisation behaviour.

Slide 23 Where Does The Design Matrix X Come From? (solved)



4A slide 23 — Ink: rows of [1 x_i] stacked into X; Xθ gives every ŷ at once.

$$\hat{y}_i = \theta_0 + \theta_1 x_i = \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}.$$

The 1 multiplies the intercept; x_i multiplies the slope. Stack the four rows:

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}}_{X \in \mathbb{R}^{4\times 2}} \underbrace{\begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}}_{\theta \in \mathbb{R}^{2\times 1}} = \underbrace{\begin{bmatrix} \theta_0 \\ \theta_0 + \theta_1 \\ \theta_0 + 2\theta_1 \\ \theta_0 + 3\theta_1 \end{bmatrix}}_{\hat{y} \in \mathbb{R}^{4\times 1}}, \qquad y = \begin{bmatrix} 2 \\ 5 \\ 6 \\ 9 \end{bmatrix}.$$

Dimensions: $(4 \times 2)(2 \times 1) = 4 \times 1$ ✓, same shape as y, so $r = y - X\theta \in \mathbb{R}^4$ ✓.

Slide 24 The Objective Is A Function Of The Parameters

$$\hat{y} = X\theta, \quad r = y - \hat{y}, \quad J(\theta) = \tfrac{1}{2}\|X\theta - y\|_2^2.$$

Once data are observed, X and y are **fixed**; the variable is θ.

Input space $x \in \mathbb{R}$	Parameter space $\theta \in \mathbb{R}^2$
the fitted rule is a **line** in x	the objective is a **bowl-shaped surface** over (θ_0, θ_1)

Each *point* in parameter space is a whole *line* in input space.

Careful. This objective is convex **in** θ, with X and y held fixed.

Slide 25 Differentiate The Objective: The Normal Equations (solved)

Expand (A7: $(X\theta)^\top = \theta^\top X^\top$, and $y^\top X\theta = \theta^\top X^\top y$ since it's a scalar):

$$J(\theta) = \tfrac{1}{2}(X\theta - y)^\top (X\theta - y) = \tfrac{1}{2}\theta^\top X^\top X\theta - \theta^\top X^\top y + \tfrac{1}{2}y^\top y.$$

Differentiate (A9; $X^\top X$ symmetric): $\nabla J(\theta) = X^\top X\theta - X^\top y = X^\top(X\theta - y)$.

Set to zero:

$$\boxed{X^\top X\,\hat{\theta} = X^\top y} \qquad (\text{the normal equations}).$$

Shapes: $X^\top X$ is 2×2, $X^\top y$ is 2×1 ✓.

Numbers:

$$X^\top X = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}, \qquad X^\top y = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix} = \begin{bmatrix} 22 \\ 44 \end{bmatrix}$$

($\sum x_i y_i = 0 + 5 + 12 + 27 = 44$.) Determinant $= 56 - 36 = 20$:

$$\hat{\theta} = \frac{1}{20}\begin{bmatrix} 14 & -6 \\ -6 & 4 \end{bmatrix}\begin{bmatrix} 22 \\ 44 \end{bmatrix} = \frac{1}{20}\begin{bmatrix} 308 - 264 \\ -132 + 176 \end{bmatrix} = \frac{1}{20}\begin{bmatrix} 44 \\ 44 \end{bmatrix} = \begin{bmatrix} \mathbf{2.2} \\ \mathbf{2.2} \end{bmatrix}.$$

Fitted line $\hat{y} = 2.2 + 2.2x$: overhead 2.2 ms, plus 2.2 ms per thousand rows.

x_i	0	1	2	3
y_i	2	5	6	9
$\hat{y}_i$	2.2	4.4	6.6	8.8
r_i	−0.2	0.6	−0.6	0.2

SSE $= 0.04 + 0.36 + 0.36 + 0.04 = 0.8$ (vs 23.5 for the slide-9 guess).

Scalar form (same answer): $\hat{\theta}_1 = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \dfrac{(-1.5)(-3.5) + (-0.5)(-0.5) + (0.5)(0.5) + (1.5)(3.5)}{2.25 + 0.25 + 0.25 + 2.25} = \dfrac{11}{5} = 2.2,\ \hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x} = 5.5 - 3.3 = 2.2.$

Slide 26 From Parameter Estimate To Projection Matrix (solved)

Columns of X linearly independent ⇒ $X^\top X$ invertible:

$$\hat{\theta} = (X^\top X)^{-1} X^\top y, \qquad \hat{y} = X\hat{\theta} = \underbrace{X(X^\top X)^{-1} X^\top}_{P_X}\, y.$$

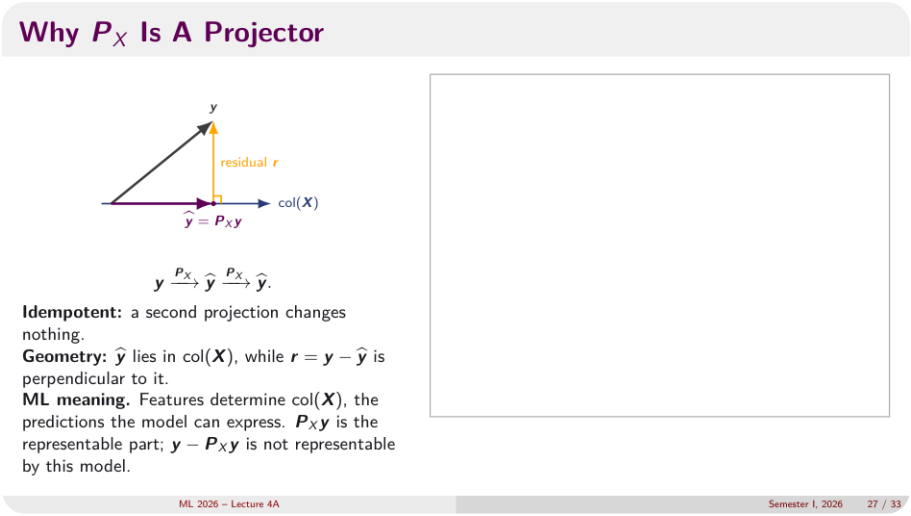
Numbers. Entry (i, j) of P_X is $[1\ x_i]\,(X^\top X)^{-1}[1\ x_j]^\top = \frac{1}{20}(14 - 6x_i - 6x_j + 4x_i x_j)$:

$$P_X = \begin{bmatrix} 0.7 & 0.4 & 0.1 & -0.2 \\ 0.4 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.4 \\ -0.2 & 0.1 & 0.4 & 0.7 \end{bmatrix}.$$

Check: row 1 times y = $1.4 + 2.0 + 0.6 - 1.8 = 2.2 = \hat{y}_1$ ✓. Symmetric ✓. Trace $= 0.7 + 0.3 + 0.3 + 0.7 = 2 = \mathrm{rank}(X)$ ✓.

P_X depends only on X (the inputs), not on y: it's a fixed machine that turns any response vector into its fitted values.

Slide 27 Why P_X Is A Projector

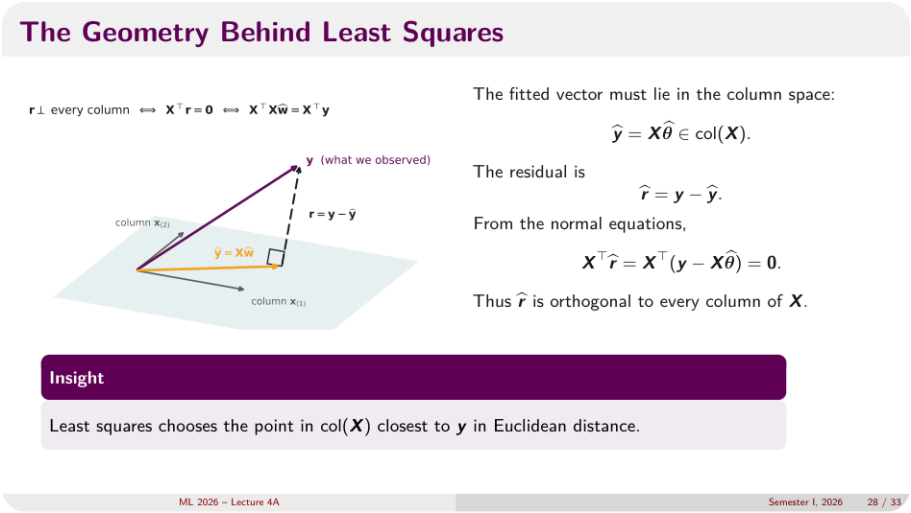


4A slide 27 — y splits into ŷ in Col(X) plus a perpendicular residual.

$y \xrightarrow{P_X} \hat{y} \xrightarrow{P_X} \hat{y}$.

- **Idempotent:** a second projection changes nothing: $P_X^2 = X(X^\top X)^{-1} \underbrace{X^\top X(X^\top X)^{-1}}_{I} X^\top = P_X$.
- **Geometry:** $\hat{y}$ lies in $\mathrm{Col}(X)$; $r = y - \hat{y}$ is perpendicular to it.
- **ML meaning:** the features determine $\mathrm{Col}(X)$, the predictions the model can express. $P_X y$ is the **representable** part of y; $y - P_X y$ is what this model **cannot represent**.

Slide 28 The Geometry Behind Least Squares



4A slide 28 — The plane is Col(X); the residual is perpendicular to it.

$\hat{y} = X\hat{\theta} \in \mathrm{Col}(X)$ and $\hat{r} = y - \hat{y}$. From the normal equations:

$$X^\top \hat{r} = X^\top (y - X\hat{\theta}) = X^\top y - X^\top X\hat{\theta} = 0.$$

So $\hat{r}$ is orthogonal to **every column** of X.

Check on our data: column of 1s: $\sum \hat{r}_i = -0.2 + 0.6 - 0.6 + 0.2 = 0$ ✓. Column x: $\sum x_i \hat{r}_i = 0 + 0.6 - 1.2 + 0.6 = 0$ ✓.

(Consequence: with an intercept, **residuals always sum to zero**, and the fitted line passes through $(\bar{x}, \bar{y}) = (1.5, 5.5)$: $2.2 + 2.2 \times 1.5 = 5.5$ ✓.)

> **Insight:** least squares picks the point in $\mathrm{Col}(X)$ **closest to** y in Euclidean distance.

Slide 29 Test Convexity In θ (solved)

$\nabla_\theta J = X^\top (X\theta - y)$. Differentiate again: $\nabla^2_\theta J = X^\top X$ (constant: doesn't depend on θ).

Test with an arbitrary v:

$$v^\top (X^\top X) v = (Xv)^\top (Xv) = \|Xv\|_2^2 \geq 0.$$

So the Hessian is PSD, and J is **convex in** θ, for any data. (A10)

Our data: $X^\top X = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$ has trace 18 and determinant 20, so eigenvalues $9 \pm \sqrt{61} \approx 16.8,\ 1.19$: both positive ⇒ **positive definite** ⇒ strictly convex ⇒ $\hat{\theta} = (2.2, 2.2)$ is the **unique** minimiser.

Slide 30 Matrix Rank Counts Independent Column Directions (solved)

$\mathrm{rank}(X) = \dim(\mathrm{Col}(X))$.

$$X_A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} : \text{ rank } 2, \qquad X_B = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} : \text{ rank } 1.$$

- X_A: is $(0,1,2,3) = c\,(1,1,1,1)$ for some c? First entry forces $c = 0$, then the second fails. Independent ⇒ rank 2 (full column rank).
- X_B: second column = first column ⇒ only one direction ⇒ rank 1 < 2. Also $\det X_B = 2 - 2 = 0$.

Slide 31 A Convex Objective Can Have A Flat Valley (solved)

$X = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. With $s = \theta_1 + \theta_2$:

$$X\theta = \begin{bmatrix} \theta_1 + \theta_2 \\ 2\theta_1 + 2\theta_2 \end{bmatrix} = \begin{bmatrix} s \\ 2s \end{bmatrix}, \qquad J(\theta) = \tfrac{1}{2}\left[(s-1)^2 + (2s-2)^2\right] = \tfrac{5}{2}(s-1)^2.$$

J depends on θ **only through** s. Minimum $J = 0$ at $s = 1$:

$$\text{every minimiser: } \{\theta : \theta_1 + \theta_2 = 1\} = \{(t,\ 1-t) : t \in \mathbb{R}\}.$$

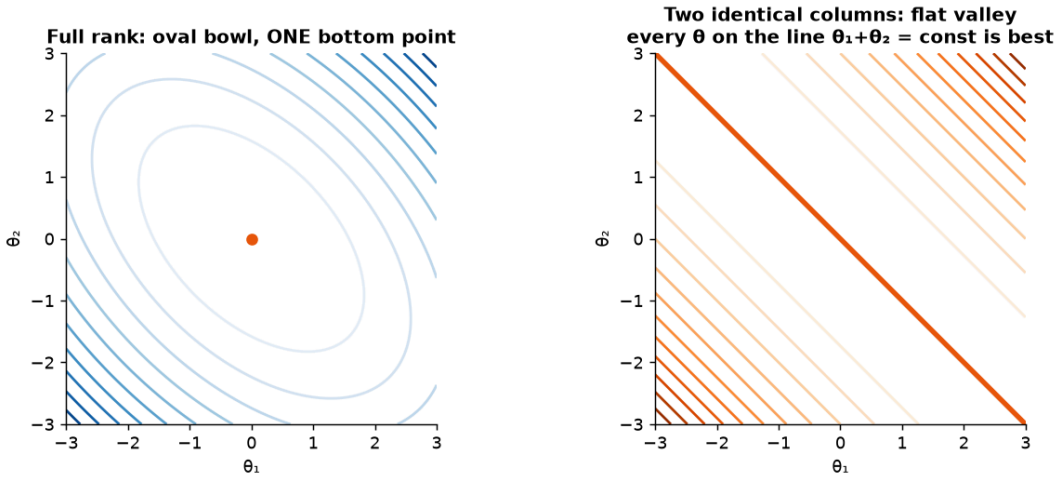
A whole line of minimisers: $(1,0)$, $(0,1)$, $(0.5, 0.5)$, $(100, -99)$... all fit perfectly. Moving along $v = (1,-1)$ changes nothing, because $Xv = 0$.

Normal equations confirm: $X^\top X = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$ (singular), $X^\top y = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$, giving the single equation $5\theta_1 + 5\theta_2 = 5$.

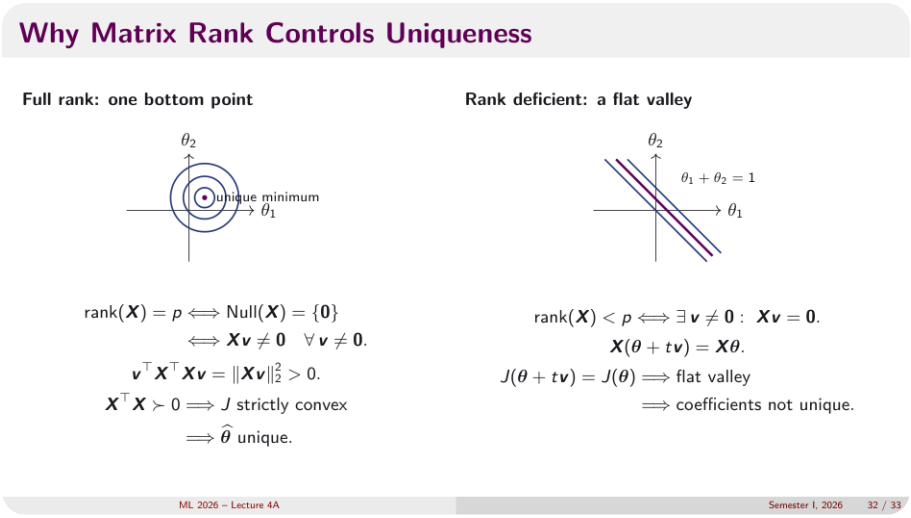
> **Insight:** convex ≠ unique. The bowl is a **trough**. (This is what perfectly correlated features do; ridge regression, L7, adds $\lambda\|\theta\|^2$ to pick the smallest-norm point $(0.5, 0.5)$.)

Slide 32

Why Matrix Rank Controls Uniqueness



Full rank gives one lowest point; two identical columns give a whole line of equally good θ.

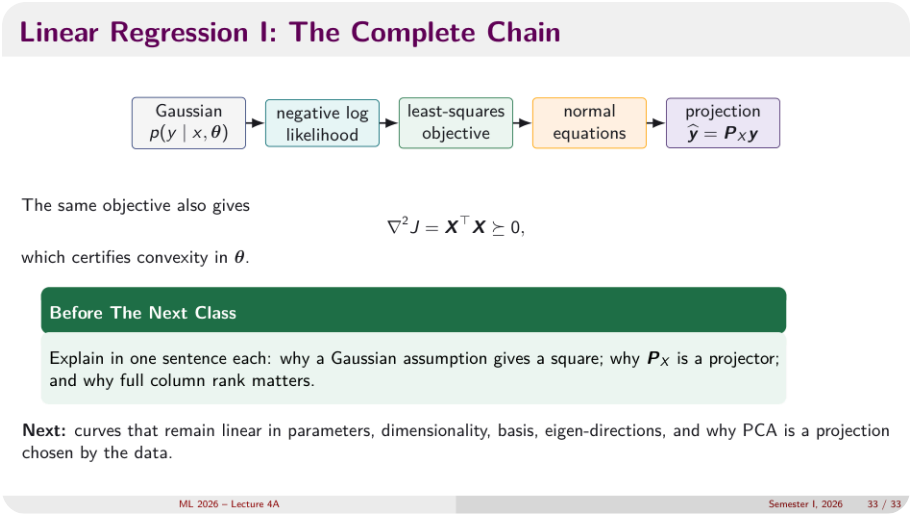


4A slide 32 — Left: rings around one point. Right: parallel lines, a flat valley.

Full rank: one bottom point	Rank deficient: a flat valley
$\text{rank}(X) = p \iff \text{Null}(X) = \{0\}$	$\text{rank}(X) < p \iff \exists v \neq 0 : Xv = 0$
$\iff Xv \neq 0$ for all $v \neq 0$	$X(\theta + tv) = X\theta$ for all t
$\Rightarrow v^\top X^\top Xv = \|Xv\|^2 > 0$	$\Rightarrow J(\theta + tv) = J(\theta)$: flat valley
$X^\top X \succ 0 \Rightarrow J$ strictly convex $\Rightarrow \hat{\theta}$ unique	coefficients **not** unique

Note: even when $\hat{\theta}$ isn't unique, the **fitted values $\hat{y}$ are** (the projection of y onto $\mathrm{Col}(X)$ is one point). Predictions are identifiable; individual coefficients aren't.

Slide 33 Linear Regression I: The Complete Chain



4A slide 33 — Gaussian → –log likelihood → least squares → normal equations → projection.

Gaussian $p(y \mid x, \theta)$ → negative log-likelihood → least-squares objective → normal equations → projection $\hat{y} = P_X y$.

The same objective gives $\nabla^2 J = X^\top X \succeq 0$, which **certifies convexity in θ**.

Before next class — one sentence each (solved):

- **Why a Gaussian gives a square:** the Gaussian density has the residual squared in its exponent, and $-\log$ brings that exponent down as $r^2/2\sigma^2$.
- **Why P_X is a projector:** $P_X^2 = P_X$ (and $P_X^\top = P_X$), so it sends y to the nearest point of $\mathrm{Col}(X)$ and leaves points already there unchanged.
- **Why full column rank matters:** it makes $X^\top X$ invertible (positive definite), so the bowl has a single bottom point and $\hat{\theta}$ is unique.

Next: curves still linear in parameters, dimensionality, basis, eigen-directions, and why PCA is a projection chosen by the data.

Part B2 Slide by Slide (all 24 slides)

4B is a **solved classroom copy**: it redoes the fit through the *algebraic* and *geometric* routes (probability and optimisation routes continue next class). Companion lab: Open the interactive sections.

Running dataset for 4B: $x = (0, 1, 2)$, $y = (1, 2, 2)$, $\phi(x) = [1, x]^\top$.

Slide 1 Title

Lecture 4B: Solved Classroom Copy Through Projection.

Slide 2 Course Roadmap

Covered: basis functions, design matrix, least squares, normal equations, perpendicular residuals, projection matrix.

Slide 3 Where Lecture 4A Stopped

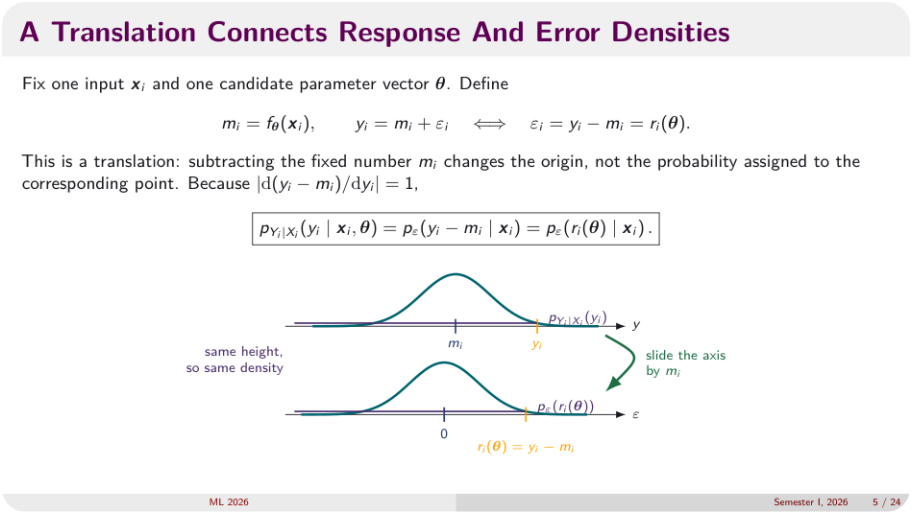
$$\hat{y}_i = f_\theta(x_i), \quad r_i(\theta) = y_i - \hat{y}_i, \quad \hat{\theta} = \arg\min_\theta \sum_{i=1}^{n} \ell(r_i(\theta)).$$

Question left in the room: if the errors aren't normal, what objective should we use? Answer: the negative log of whatever the error density is (slides 5–6).

Slide 4 Learning Objectives

- map an error density to a loss via negative log-likelihood;
- build a design matrix from basis functions;
- write least squares as a nearest-point problem in $\mathrm{Col}(X)$;
- derive the normal equations and explain why the residual is perpendicular to the model space;
- read $P_X y$ and $(I - P_X)y$ as the representable and unrepresentable parts of y.

Slide 5 A Translation Connects Response And Error Densities



4B slide 5 — Sliding the noise bell by fθ(x) gives the density of y.

Fix x_i and a candidate θ. Let $m_i = f_\theta(x_i)$. Then

$$y_i = m_i + \varepsilon_i \iff \varepsilon_i = y_i - m_i = r_i(\theta).$$

Subtracting a fixed number shifts the origin; it doesn't change heights. The Jacobian $|d(y_i - m_i)/dy_i| = 1$, so

$$p_{Y_i|X_i}(y_i \mid x_i, \theta) = p_\varepsilon(r_i(\theta) \mid x_i).$$

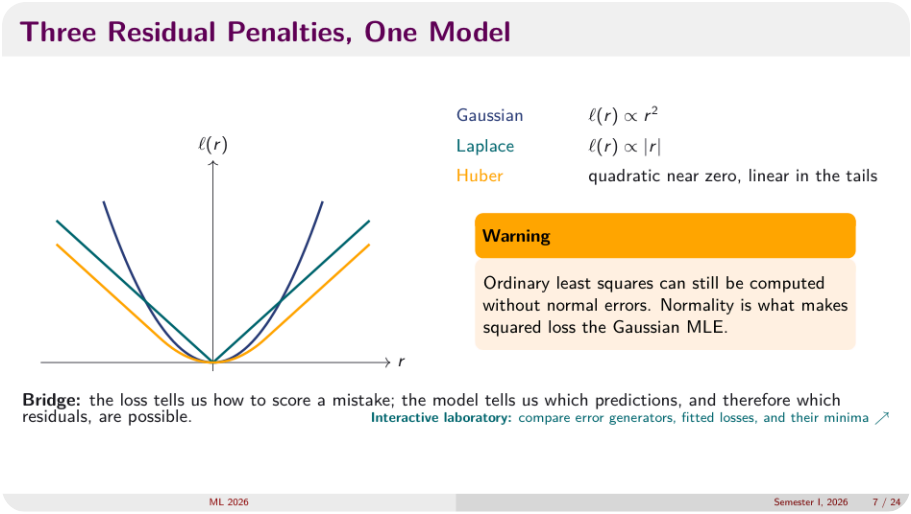
Picture: the response density centred at m_i, slid left by m_i, is the error density centred at 0. Same height at y_i and at r_i.

Slide 6 From One Residual Density To A Loss

$$p(y \mid X, \theta) = \prod_{i=1}^{n} p_\varepsilon(r_i(\theta) \mid x_i) \;\Rightarrow\; \hat{\theta}_{\mathrm{MLE}} = \arg\min_\theta \sum_{i=1}^{n} \underbrace{-\log p_\varepsilon(r_i(\theta) \mid x_i)}_{\ell_i(\theta)}\,.$$

Summary: the error model chooses the likelihood-shaped loss. (A4)

Slide 7 Three Residual Penalties, One Model



4B slide 7 — Gaussian (parabola), Laplace (V), Huber (both).

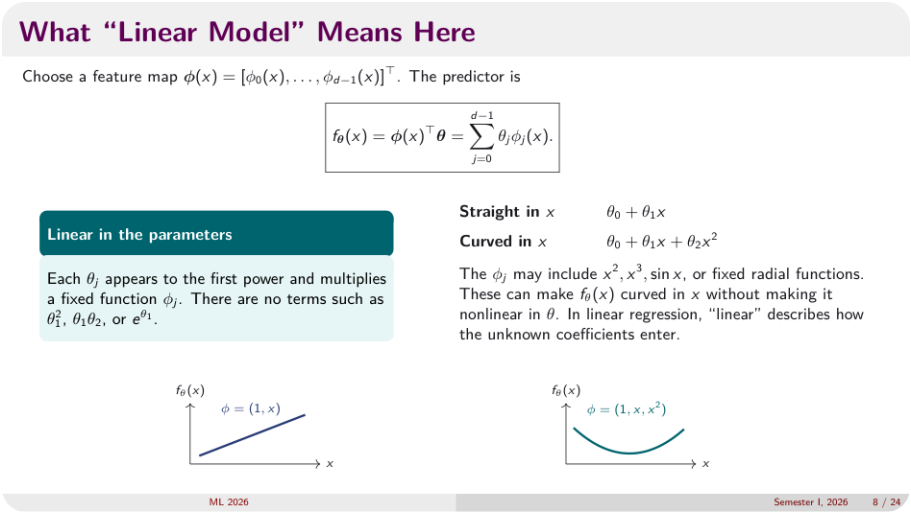
Plot of $\ell(r)$: Gaussian $\propto r^2$ (parabola), Laplace $\propto |r|$ (V shape), Huber (parabola near 0, straight lines in the tails).

Careful. OLS can still be **computed** without normal errors (it's just a minimisation). Normality is what makes squared loss **the Gaussian MLE**.

Bridge: the loss says how to score a mistake; the model says which predictions, and so which residuals, are possible.

Slide 8

What "Linear Model" Means Here



4B slide 8 — 'Linear' = linear in θ; the right plot is a curve in x yet still linear regression.

$$f_\theta(x) = \phi(x)^\top \theta = \sum_{j=0}^{d-1} \theta_j \phi_j(x).$$

Linear in the parameters: each θ_j appears to the first power and multiplies a fixed function ϕ_j. No θ_1^2, $\theta_1\theta_2$, e^{θ_1}.

Features	Model	Shape in x
$\phi = (1, x)$	$\theta_0 + \theta_1 x$	straight
$\phi = (1, x, x^2)$	$\theta_0 + \theta_1 x + \theta_2 x^2$	curved

ϕ_j may be x^2, x^3, $\sin x$, fixed radial bumps... "Linear" describes how the **unknown coefficients** enter.

Slide 9 Stack The Rows: The Design Matrix Appears

Stack The Rows: The Design Matrix Appears

Use one dataset throughout the geometric argument:

$$x = (0, 1, 2), \qquad \boldsymbol{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \qquad \phi(x) = \begin{bmatrix} 1 \\ x \end{bmatrix}.$$

For observation i,

$$\widehat{y}_i = \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}.$$

Stack the three rows. Then multiply the resulting matrix by $\boldsymbol{\theta}$ and check every dimension.

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}}_{X \in \mathbb{R}^{3\times 2}} \underbrace{\begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}}_{\boldsymbol{\theta} \in \mathbb{R}^2} = \underbrace{\begin{bmatrix} \theta_0 \\ \theta_0 + \theta_1 \\ \theta_0 + 2\theta_1 \end{bmatrix}}_{\widehat{\boldsymbol{y}} = X\boldsymbol{\theta} \in \mathbb{R}^3}.$$

$$\boldsymbol{r} = \boldsymbol{y} - X\boldsymbol{\theta} \in \mathbb{R}^3.$$

Each *row* builds one prediction. Each *column* records one feature pattern across all observations.

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4B slide 9 — Each row is one observation; each column one feature.

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}}_{X \in \mathbb{R}^{3\times 2}} \underbrace{\begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}}_{\theta \in \mathbb{R}^2} = \underbrace{\begin{bmatrix} \theta_0 \\ \theta_0 + \theta_1 \\ \theta_0 + 2\theta_1 \end{bmatrix}}_{\hat{y} = X\theta \in \mathbb{R}^3}, \qquad r = y - X\theta \in \mathbb{R}^3.$$

Each **row** builds one prediction; each **column** records one feature pattern across all observations.

Slide 10 What Does "Span" Mean?

What Does "Span" Mean?

For the running matrix, name its columns:

$$X = [\boldsymbol{c}_0 \ \boldsymbol{c}_1], \qquad \boldsymbol{c}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{c}_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

Every parameter choice produces

$$X\boldsymbol{\theta} = \theta_0 \boldsymbol{c}_0 + \theta_1 \boldsymbol{c}_1.$$

Compute the predictions for $\boldsymbol{\theta} = (1, 0)$, $(0, 1)$, and $(1, \frac{1}{2})$. What remains fixed, and what is allowed to vary?

$$X \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \boldsymbol{c}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad X \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \boldsymbol{c}_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix},$$

$$X \begin{bmatrix} 1 \\ 1/2 \end{bmatrix} = \boldsymbol{c}_0 + \tfrac{1}{2}\boldsymbol{c}_1 = \begin{bmatrix} 1 \\ 3/2 \\ 2 \end{bmatrix}.$$

The columns stay fixed; the coefficients vary. The **span** is simply the set of *all* vectors obtainable as $a\boldsymbol{c}_0 + b\boldsymbol{c}_1$ for real numbers a, b.

$$\mathrm{Col}(X) = \mathrm{span}\{\boldsymbol{c}_0, \boldsymbol{c}_1\}.$$

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4B slide 10 — Span = everything you can build from the columns.

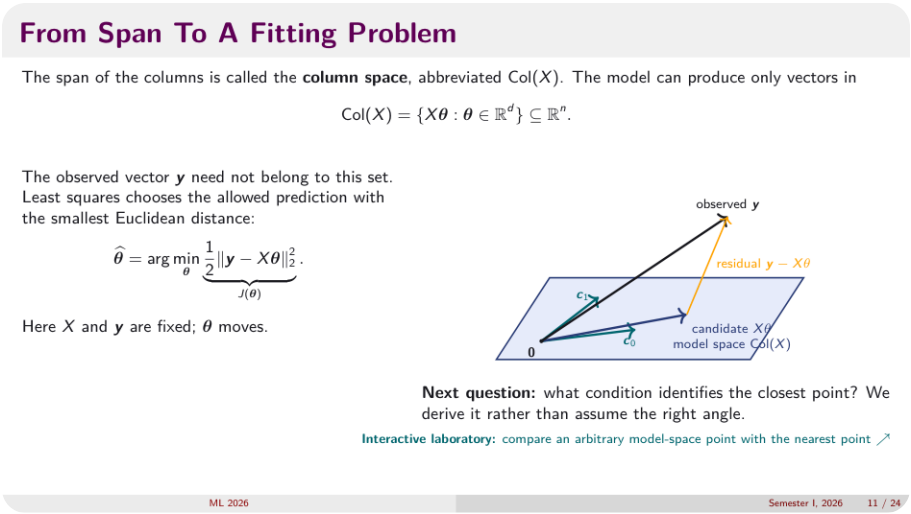
$X = [c_0\ c_1]$, $c_0 = (1,1,1)$, $c_1 = (0,1,2)$, and $X\theta = \theta_0 c_0 + \theta_1 c_1$.

θ	$X\theta$
$(1,0)$	$c_0 = (1,1,1)$
$(0,1)$	$c_1 = (0,1,2)$
$(1,\frac{1}{2})$	$c_0 + \frac{1}{2}c_1 = (1,\ 1.5,\ 2)$

The columns stay fixed; the coefficients vary. **Span** = all $a\,c_0 + b\,c_1$. $\mathrm{Col}(X) = \mathrm{span}\{c_0, c_1\}$: a 2-D plane through the origin inside $\mathbb{R}^3$.

Slide 11

From Span To A Fitting Problem



4B slide 11 — y sits off the plane; the fit is the closest point on it.

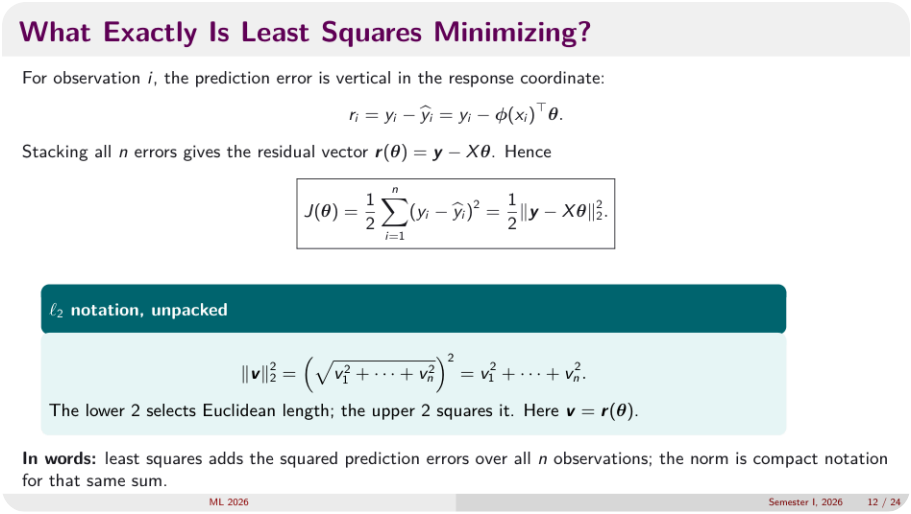
$\mathrm{Col}(X) = \{X\theta : \theta \in \mathbb{R}^d\} \subseteq \mathbb{R}^n$. The observed y needn't lie in it. (Here: is $(1,2,2) = a(1,1,1) + b(0,1,2)$? First entry: $a = 1$. Second: $b = 1$. Third: $1 + 2 = 3 \neq 2$. **No.**)

So least squares picks the allowed prediction nearest to y:

$$\hat{\theta} = \arg\min_{\theta} \underbrace{\tfrac{1}{2}\|y - X\theta\|_2^2}_{J(\theta)}, \qquad X, y \text{ fixed};\ \theta \text{ moves}.$$

Next: derive (not assume) the right angle that identifies the closest point.

Slide 12 What Exactly Is Least Squares Minimizing?



4B slide 12 — The sum of squared residuals is a squared length $\|r\|^2$.

$r_i = y_i - \phi(x_i)^\top \theta$ is the **vertical** error in the response coordinate. Stacked: $r(\theta) = y - X\theta$, and

$$J(\theta) = \tfrac{1}{2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \tfrac{1}{2}\|y - X\theta\|_2^2, \qquad \|v\|_2^2 = \left(\sqrt{v_1^2 + \cdots + v_n^2}\right)^2 = v_1^2 + \cdots + v_n^2.$$

The norm is just compact notation for the sum of squared errors.

Slide 13 From Least Squares To A Nearest-Point Problem

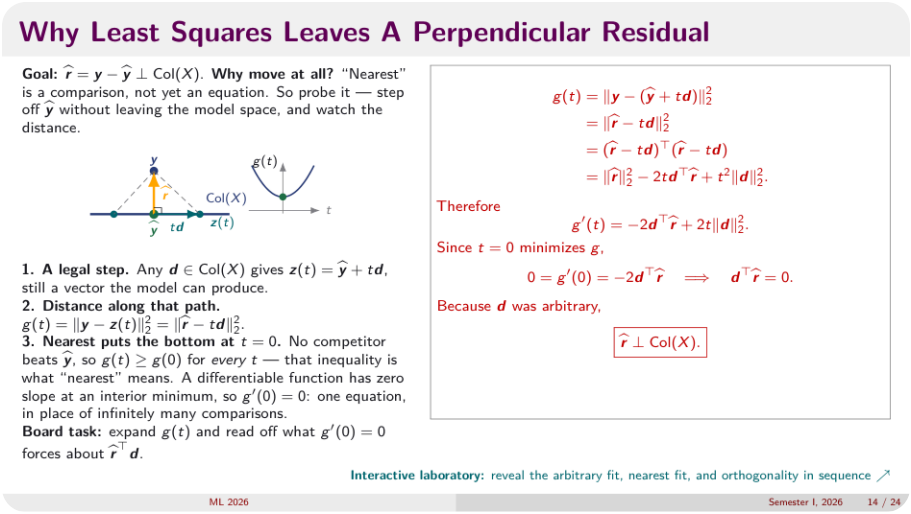
Every possible fitted vector is $z = X\theta \in \mathrm{Col}(X)$, so

$$\hat{y} = \arg\min_{z \in \mathrm{Col}(X)} \|y - z\|_2^2.$$

Among all prediction vectors the model can produce, choose the one closest to y.

Careful. We square the **vertical** response errors, not each point's perpendicular distance to the plotted line (that would be total least squares / PCA, L5). The "perpendicular" in least squares lives in n-dimensional **observation space**, not in the x–y scatter plot.

Slide 14 Why Least Squares Leaves A Perpendicular Residual



4B slide 14 — Any other point on the plane is farther (Pythagoras).

Goal: $\hat{r} = y - \hat{y} \perp \mathrm{Col}(X)$. "Nearest" is a comparison, not yet an equation, so probe it:

1. **A legal step:** for any $d \in \mathrm{Col}(X)$, $z(t) = \hat{y} + td$ is still producible by the model.
2. **Distance along the path:** $g(t) = \|y - z(t)\|^2 = \|\hat{r} - td\|^2$.
3. **Nearest ⇒ bottom at $t = 0$:** $g(t) \geq g(0)$ for all t, so $g'(0) = 0$. One equation instead of infinitely many comparisons.

Expand:

$$g(t) = (\hat{r} - td)^\top (\hat{r} - td) = \|\hat{r}\|^2 - 2t\, d^\top \hat{r} + t^2 \|d\|^2, \qquad g'(t) = -2d^\top \hat{r} + 2t\|d\|^2.$$

$$0 = g'(0) = -2d^\top \hat{r} \;\Rightarrow\; d^\top \hat{r} = 0.$$

Since d was any vector in $\mathrm{Col}(X)$: $\hat{r} \perp \mathrm{Col}(X)$.

Slide 15 Route 1: Algebra Solves The Normal Equations

1. Expand: $J(\theta) = \frac{1}{2} y^\top y - \theta^\top X^\top y + \frac{1}{2} \theta^\top X^\top X \theta$.
2. Differentiate: $\nabla J(\theta) = X^\top (X\theta - y)$.
3. Set to zero: $X^\top X \hat{\theta} = X^\top y$.

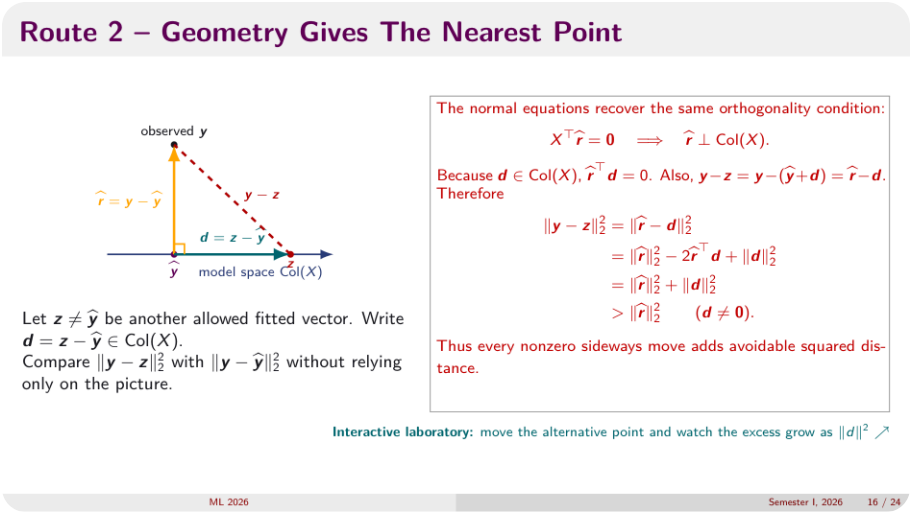
4. Name: the **normal equations**. Gradient and both sides are in $\mathbb{R}^d$.

Numbers on the 4B data (not on the slide, worked here):

$$X^\top X = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \quad X^\top y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}, \quad \det = 6, \quad \hat{\theta} = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} 5 \\ 6 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 7 \\ 3 \end{bmatrix} = \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix}.$$

$\hat{y} = (7/6,\ 5/3,\ 13/6)$, $\hat{r} = (-1/6,\ 1/3,\ -1/6)$. Check $X^\top \hat{r}$: $-\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = 0$ ✓ and $0 + \frac{1}{3} - \frac{1}{3} = 0$ ✓. SSE $= \|\hat{r}\|^2 = \frac{1+4+1}{36} = \frac{1}{6}$.

Slide 16 Route 2: Geometry Gives The Nearest Point



4B slide 16 — Route 2: drop a perpendicular, no calculus needed.

Let $z \neq \hat{y}$ be any other allowed fit, $d = z - \hat{y} \in \mathrm{Col}(X)$. From the normal equations $X^\top \hat{r} = 0$, so $\hat{r}^\top d = 0$. And $y - z = \hat{r} - d$:

$$\|y - z\|^2 = \|\hat{r}\|^2 - 2\hat{r}^\top d + \|d\|^2 = \|\hat{r}\|^2 + \|d\|^2 > \|\hat{r}\|^2 \quad (d \neq 0).$$

Every nonzero sideways move adds exactly $\|d\|^2$ of avoidable squared distance (Pythagoras). So $\hat{y}$ is *the* nearest point, and it's unique.

Slide 14 showed nearest ⇒ perpendicular; this slide shows perpendicular ⇒ nearest.

Slide 17 From The Parameter Estimate To The Projection Matrix

$X^\top X$ invertible (independent columns):

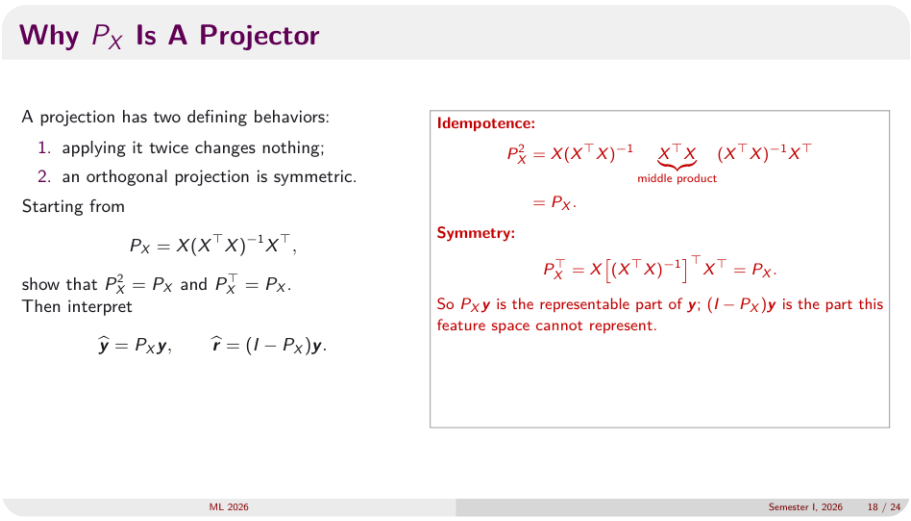
$$\hat{\theta} = (X^\top X)^{-1} X^\top y, \qquad \hat{y} = X(X^\top X)^{-1} X^\top y = P_X y, \qquad P_X = X(X^\top X)^{-1} X^\top.$$

P_X is the orthogonal projection onto $\mathrm{Col}(X)$.

Numbers: entry $(i,j) = \frac{1}{6}(5 - 3x_i - 3x_j + 3x_i x_j)$:

$$P_X = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}, \qquad P_X y = \frac{1}{6}\begin{bmatrix} 5+4-2 \\ 2+4+4 \\ -1+4+10 \end{bmatrix} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix} = \hat{y}\ \checkmark.$$

Slide 18 Why P_X Is A Projector



4B slide 18 — P² = P (projecting twice changes nothing) and Pᵀ = P.

Idempotence:

$$P_X^2 = X(X^\top X)^{-1} \underbrace{X^\top X(X^\top X)^{-1}}_{=I} X^\top = P_X.$$

Symmetry: $P_X^\top = (X^\top)^\top \left[(X^\top X)^{-1}\right]^\top X^\top = X(X^\top X)^{-1}X^\top = P_X$ (the inverse of a symmetric matrix is symmetric).

$$\hat{y} = P_X y \text{ (representable part)}, \qquad \hat{r} = (I - P_X) y \text{ (what this feature space can't represent)}.$$

Numbers: $I - P_X = \frac{1}{6}\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix} = \frac{1}{6} u u^\top$ with $u = (1, -2, 1)$. u is perpendicular to both c_0 ($1 - 2 + 1 = 0$) and c_1 ($0 - 2 + 2 = 0$): it's the one direction in $\mathbb{R}^3$ the model can't reach. $(I - P_X) y = \frac{1}{6}(u^\top y)\, u = \frac{-1}{6}(1, -2, 1) = \hat{r}$ ✓.

Slide 19 Class Question 1: Where Does d Come From?

1. Chosen fit and any competitor: $\hat{y} = X\hat{\theta}$, $z = X\theta'$. Both in $\mathrm{Col}(X)$.
2. Subtract: $d := z - \hat{y} = X(\theta' - \hat{\theta}) \in \mathrm{Col}(X)$, and $z = \hat{y} + d$.
3. Introduce a scalar only to move along it: $z(t) = \hat{y} + td$, $z(0) = \hat{y}$, $z(1) = z$.

d is **not** a new model parameter. It's just "competitor minus chosen fit" in observation space.

Slide 20 Class Question 1: Which Residual Depends On d?

- Base-point residual $\hat{r} := y - \hat{y}$: fixed once $\hat{y}$ is chosen.
- Moving residual: $r(t; d) := y - z(t) = (y - \hat{y}) - td = \hat{r} - td$.

The moving residual depends on t and d; at $t = 0$, $r(0; d) = \hat{r}$ for every direction.

Slide 21 Class Question 1: Differentiate Along That Direction

$$g(t) := \tfrac{1}{2}\|\hat{r} - td\|^2 = \tfrac{1}{2}\hat{r}^\top \hat{r} - t\,\hat{r}^\top d + \tfrac{1}{2}t^2 d^\top d, \qquad g'(t) = -\hat{r}^\top d + t\, d^\top d.$$

$$g'(0) = -\hat{r}^\top d = 0 \quad \text{for every } d \in \mathrm{Col}(X) \;\Rightarrow\; \hat{r} \perp \mathrm{Col}(X).$$

d is held fixed while differentiating in t; that's why it remains in the derivative. (Same as slide 14, with a ½.)

Slide 22 Class Question 2: Expand The Transpose Before Differentiating

1. Transpose: $(y - X\theta)^\top = y^\top - \theta^\top X^\top$.
2. Multiply in order: $J = \frac{1}{2}(y^\top y - y^\top X\theta - \theta^\top X^\top y + \theta^\top X^\top X\theta)$.
3. $y^\top X\theta$ is 1×1, so it equals its transpose $\theta^\top X^\top y$. Combine:

$$J(\theta) = \tfrac{1}{2}y^\top y - \theta^\top X^\top y + \tfrac{1}{2}\theta^\top (X^\top X)\theta.$$

Slide 23 Class Question 2: Derive The Quadratic Rule

$A = X^\top X$, $q(\theta) = \frac{1}{2}\theta^\top A\theta = \frac{1}{2}\sum_j \sum_k \theta_j A_{jk}\theta_k$.

$$\frac{\partial q}{\partial \theta_m} = \tfrac{1}{2}\sum_k A_{mk}\theta_k + \tfrac{1}{2}\sum_j \theta_j A_{jm} = \left[\tfrac{1}{2}(A + A^\top)\theta\right]_m \Rightarrow \nabla_\theta q = \tfrac{1}{2}(A + A^\top)\theta = A\theta \ (A \text{ symmetric}).$$

With $\nabla_\theta(\theta^\top X^\top y) = X^\top y$:

$$\nabla_\theta J = -X^\top y + X^\top X\theta = X^\top (X\theta - y).$$

Slide 24 Class Question 2: Check The Answer With Differentials

$r = y - X\theta$, $J = \frac{1}{2}r^\top r$, $dr = -X\, d\theta$.

$$dJ = \tfrac{1}{2}\left[(dr)^\top r + r^\top dr\right] = r^\top dr = -r^\top X\, d\theta = (-X^\top r)^\top d\theta.$$

Since $dJ = (\nabla_\theta J)^\top d\theta$: $\nabla_\theta J = -X^\top r = X^\top (X\theta - y)$ ✓.

Dimension check: $X^\top \in \mathbb{R}^{d\times n}$, $r \in \mathbb{R}^n \Rightarrow X^\top r \in \mathbb{R}^d$, matching $\theta \in \mathbb{R}^d$.

Part C Practice

1. For the 4A data, compute the SSE of the line $\hat{y} = 2 + 2x$. Is it more or less than 0.8? Why must it be?
2. A model predicts $\hat{y} = (3, 5, 7)$ for $y = (4, 4, 9)$. Compute SSE, MSE, RMSE. Which one is in the units of y?
3. Fit a constant a to $1, 2, 2, 3, 12$ under squared and under absolute loss. Which is pulled by the 12?
4. Fit a constant to $2, 3, 4, 5, 20$ under Huber loss with $\delta = 1$ and with $\delta = 3$. (Hint: the minimiser solves $\sum_i \psi(y_i - a) = 0$ where $\psi(r) = r$ for $|r| \le \delta$ and $\pm\delta$ otherwise.)
5. Errors follow $p(\varepsilon) \propto e^{-\varepsilon^4}$. What loss does MLE give? Would it be more or less sensitive to outliers than squared loss?
6. Is each model linear regression? (a) $\theta_0 + \theta_1 \sin x$ (b) $\theta_0 + e^{\theta_1 x}$ (c) $\theta_0 + \theta_1 x + \theta_2 x \log x$ (d) $\theta_0 \theta_1 x$.
7. Write X for a quadratic fit $\phi = (1, x, x^2)$ at $x = (-1, 0, 1, 2)$. What's its shape? Its rank?
8. Fit a line through the origin, $\hat{y} = \theta x$, by least squares. Derive $\hat{\theta}$ in scalar form. Evaluate it on the 4A data.
9. For the 4B data, verify $P_X^2 = P_X$ by multiplying out row 1 of P_X times column 1 of P_X.
0. Show that if X has a column of ones, the least-squares residuals sum to zero.
1. $X = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$. Rank? Is the least-squares $\hat{\theta}$ unique? Is $\hat{y}$ unique?
2. A colleague says "least squares is convex, so the coefficients are unique." Correct them in one sentence.
3. Why is $\operatorname{trace}(P_X) = d$? (Hint: $\operatorname{trace}(AB) = \operatorname{trace}(BA)$.)
4. Explain the difference between ε_i and $\hat{r}_i$. Can $\sum \hat{r}_i = 0$ while $\sum \varepsilon_i \neq 0$?

1. $\hat{y} = (2, 4, 6, 8)$, $r = (0, 1, 0, 1)$, SSE $= 2 > 0.8$. It must be ≥ 0.8 because $(2.2, 2.2)$ is the unique minimiser of SSE.
2. $r = (1, -1, 2)$. SSE $= 6$, MSE $= 2$, RMSE $= \sqrt{2} \approx 1.41$. RMSE is in the units of y.
3. Squared: mean $= 20/5 = 4$. Absolute: median $= 2$. The mean is pulled by the 12 (without it, both would be 2).
4. $\delta = 1$: at $a = 4$, residuals $(-2, -1, 0, 1, 16)$, $\psi = (-1, -1, 0, 1, 1)$, sum 0, so $a = 4$. $\delta = 3$: try $a = 4.25$: residuals $(-2.25, -1.25, -0.25, 0.75, 15.75)$, $\psi = (-2.25, -1.25, -0.25, 0.75, 3)$, sum $= 0$, so $a = 4.25$. As δ grows Huber moves toward the mean (6.8); as $\delta \to 0$ toward the median (4).
5. $-\log p = \varepsilon^4 +$ const, so loss r^4. **More** sensitive to outliers: a residual of 10 costs 10 000.

6. (a) yes ($\phi = (1, \sin x)$). (b) no (θ_1 inside an exponential). (c) yes ($\phi = (1, x, x\log x)$). (d) no as written ($\theta_0\theta_1$ is a product of parameters); it's equivalent to the linear βx with $\beta = \theta_0\theta_1$, but then θ_0, θ_1 themselves aren't identifiable.
7. $X = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}$, 4×3, rank 3 (four distinct x's ≥ 3 parameters, so the columns are independent).
8. $J = \frac{1}{2}\sum(y_i - \theta x_i)^2$, $J' = -\sum x_i(y_i - \theta x_i) = 0 \Rightarrow \hat\theta = \dfrac{\sum x_i y_i}{\sum x_i^2}$. On 4A: $44/14 = 3.14$. (Steeper than 2.2 because, without an intercept, the line must pass through the origin.)
9. Row 1 · column 1 $= \frac{1}{36}(5\cdot 5 + 2\cdot 2 + (-1)(-1)) = \frac{30}{36} = \frac{5}{6} = (P_X)_{11}$ ✓.
10. Normal equations ⇒ $X^\top \hat r = 0$. The row of $X^\top$ corresponding to the ones column gives $\mathbf{1}^\top \hat r = \sum_i \hat r_i = 0$.
11. Rank 1 (column 2 = 2 × column 1). $\hat\theta$ not unique: any θ with $\theta_0 + 2\theta_1 = \bar y$ works. $\hat y$ is unique: $\hat y = \bar y\,\mathbf{1}$ (projection onto the single direction $(1,1,1)$).
12. Convex only guarantees every stationary point is a global minimum; uniqueness needs **strict** convexity, i.e. full column rank of X ($X^\top X \succ 0$).
13. $\operatorname{trace}(X(X^\top X)^{-1}X^\top) = \operatorname{trace}((X^\top X)^{-1}X^\top X) = \operatorname{trace}(I_d) = d$.
14. ε_i is the unobserved random noise in the true model; $\hat r_i$ is computed after fitting. Yes: with an intercept $\sum \hat r_i = 0$ always (by construction of the fit), while the true noise has $\mathbb{E}[\varepsilon_i] = 0$ but its realised sum is almost never exactly 0.